# Family Structure and Youth Employment Instability: Evidence on Job Quality and Match Stability *

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**Abstract**

This paper studies how family structure during adolescence is related to employment instability in young adulthood, conditional on observed human capital and household financial resources. Using weekly labor market histories from the NLSY97, I compare workers who lived with a single parent at age 16 with those who lived with two resident parent figures. Data analyses establish three empirical findings: (i) workers from single-parent households experience higher post-school unemployment rate, driven primarily by more frequent separations from employment rather than slower job-finding rates; (ii) workers from single-parent households experience less favorable job-in-wage quality persistently, especially following unemployment; (iii) subsequent higher separation risk of workers from single-parent households remains, even after standardizing wage quality among direct job-to-job transitions. I interpret these facts through a job-ladder model with heterogeneous wage quality, nonwage match value, and latent match stability. In the model, family structure shifts the job-offer distributions of wage quality and match stability. The estimated model attributes most of the aggregate employment-instability gap to reduced access to stable opportunities during employed search. Consumption-equivalent welfare analysis suggests that employment instability of workers from single-parent households is equivalent to permanent 5.8% less consumption compared to workers from two-parent households. ASEP's public-sector hiring policy in Greece and WIOA Youth Program in Prince George's county in Maryland are discussed as examples of existing labor policy targeting workers from single-parent households.

**JEL Codes:** E24, J12, J63, J64.

**Keywords:** Labor Market Inequality, Family Structure, Youth Unemployment, Job Ladder, Match Stability, Labor Policy

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# 1 Introduction

Research on labor market inequality among young workers often focuses on differences in employment, unemployment, and wages or earnings. These outcomes, however, summarize the cumulative effects of underlying labor market transitions. Educational attainment is an important source of heterogeneity in these outcomes, but even among workers with similar observed human capital, labor market trajectories may differ substantially in the frequency and characteristics of transitions across jobs and labor market states. Two workers with similar job-finding rates can experience very different employment trajectories if one repeatedly moves from employment into unemployment, enters less durable matches, or reaches lower quality jobs after an unemployment spell. A flow-based perspective is therefore particularly useful in early career, when employer changes are frequent and small differences in employment continuity can cumulate into persistent differences in experience and earnings.

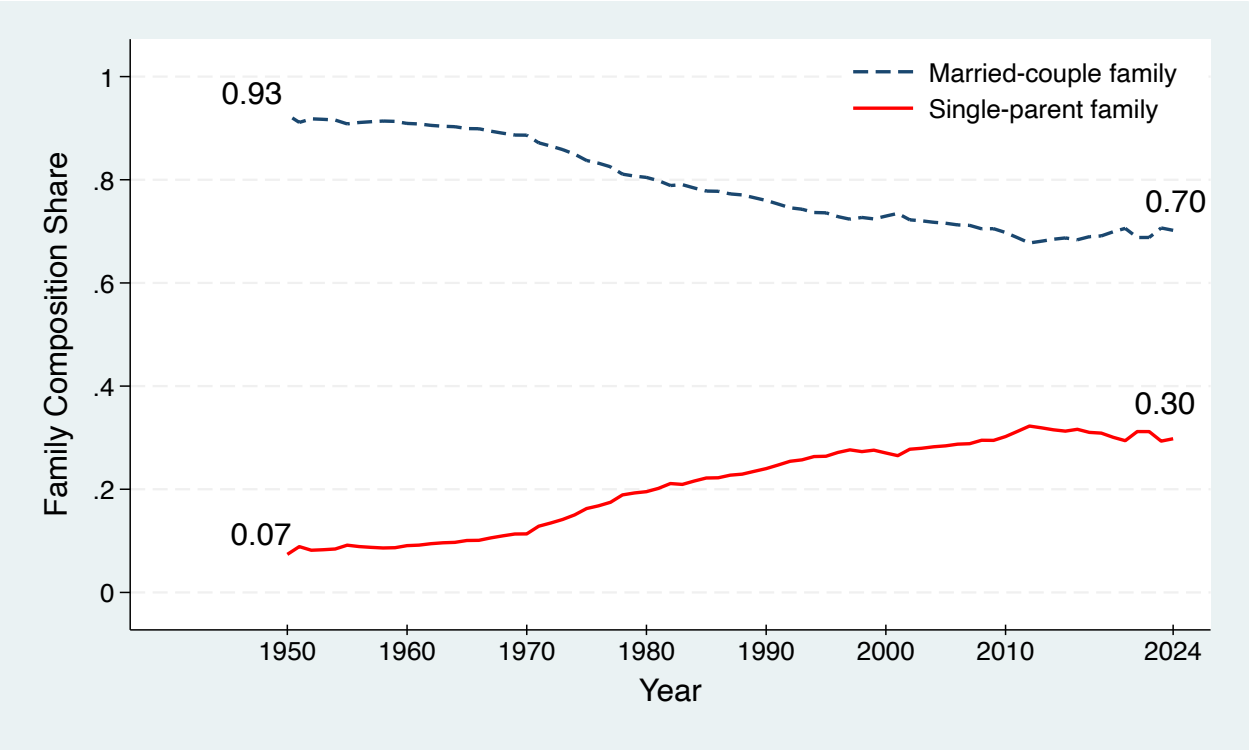
This paper asks whether family structure during adolescence is systematically associated with these early-career ladder labor dynamics. Family structure is quantitatively important and relevant because family living arrangements have changed substantially in the United States. As shown in Figure 1, among families with own children under age 18, the single parent share increased from roughly 7 percent in 1950 to about 30 percent in 2024. While this secular change does not by itself imply a causal effect of family structure on adult outcomes, it makes important to understand whether family structure is associated with the way young adults enter jobs, move between employers, and experience unemployment.

Family structure may be correlated with many dimensions of the childhood environment and may itself shape human capital accumulation. Accordingly, the empirical analysis is designed to measure *conditional associations*: I ask whether family structure differences in labor market instability remain after adjusting for observed education and other background characteristics, and which labor market frictions account for those remaining differences.

I use the National Longitudinal Survey of Youth 1997 (NLSY97). I compare respondents who lived with a single parent at age 16 with those who lived with two resident parent figures. Family structure at age 16 is measured before the post-school labor market histories studied in the paper. Family structure is defined from the household roster around at age 16. Throughout the analysis, I control for demographics, own educational attainment, parental education, and in richer specifications, measures of family and household resources.

Workers' weekly employment histories allow me to distinguish unemployment stocks from the flows that generate them. I construct post-school labor market states at the weekly frequency, identify worker-employer spells and direct employer-to-employer (EE) moves, and measure the wage quality of accepted jobs using start-year-specific wage terciles. Note that, because the NLSY97 observes accepted jobs but not rejected offers, the empirical analysis is deliberately stated in terms of the composition of *accepted* worker-employer matches rather than the firm-side offer distribu-

Figure 1: Family Composition among U.S. Families with Own Children under Age 18, 1950–2024




*Notes:* The figure reports the shares of married-couple (blue dashed line) and single-parent families (red solid line) among U.S. families with at least one own child under age 18. Single-parent families combine mother-only and father-only families. The denominator is the total number of families with own children under age 18, rather than all U.S. families or households. "Own children" include biological, step, and adopted children. The shares are calculated from the underlying family counts.
*Source:* Author's calculations using the U.S. Census Bureau, Current Population Survey, March and Annual Social and Economic Supplements, Historical Families Table FM-1.

tion.

The empirical findings are threefold. First, workers from single parent households have higher unemployment after school exit even after adjustment for observed human capital and family resources. The difference is not primarily driven by a job-finding rates. Unemployment-to-employment (UE) transition rates are similar across the two groups, whereas the employment-to-unemployment (EU) hazard is significantly higher for workers from single parent households.

Second, workers from single-parent households experience persistent disadvantages in job quality, beginning with their first jobs and continuing throughout their subsequent employment histories. To examine these differences, I compare the wage quality of accepted jobs across family structures. Workers from single-parent households are 2.11 percentage points more likely to enter the low-wage tercile and 1.87 percentage points less likely to enter the high-wage tercile in the fully controlled specification. I then distinguish jobs reached from unemployment from those reached through direct employment-to-employment (EE) transitions to identify where these differences emerge. The differences are substantially larger among reached from unemployment,

whereas wage-quality differences among jobs reached through EE transitions are close to zero. Importantly, however, the less favorable wage composition of jobs reached from unemployment is not accompanied by a corresponding difference in their subsequent employment stability.

Third, family-structure differences in subsequent employment stability emerge primarily among jobs reached through direct EE transitions, rather than among jobs entered from unemployment. After accounting for observed wage quality and other covariates, workers from single-parent households have a 14.5% higher EU hazard than workers from two-parent households among jobs reached through direct EE transitions. In contrast, among jobs entered from unemployment, the difference in EU hazards is close to zero. This pattern contrasts with the wage-quality results: family-structure differences in accepted job quality are concentrated among jobs reached from unemployment, while differences in subsequent employment stability are concentrated among jobs reached through direct EE transitions.

In other words, employment instability that workers from single-parent households face when they find jobs is two-dimensional. When they find a job from unemployment status, the wage quality of job is less favorable on average. Once they get employed and try to change employers, the subsequent job is less durable, resulting in their falling back to unemployment status again. This pattern implies that employment instability of workers from single-parent households is intertwined between the wage quality wedge and the stability wedge.

I use a quantitative job-ladder model to organize these facts and to separate two distinct margins of job access. The model builds on the heterogeneous match framework of Baydur and Mukoyama (2020). A match has three components: observed match quality $q$, nonwage match value $x$, and latent stability type $s$. Family structure enters the model only through differences in the *job offer distribution*. The first wedge, $\alpha_U$, shifts the wage quality distribution of job offer accessible from unemployment. A negative $\alpha_U$ tilts the single parent distribution toward lower wage quality. The second wedge, $\gamma_E$, shifts the stability composition of job offer distribution during employed search. A negative $\gamma_E$ lowers the probability that a job offer while on-the-job search to a worker from single parent household is of the stable type. I do not interpret either wedge as a literal firm-side offer distribution. Rather, each summarizes the opportunities that are effectively accessible to a worker after unmodeled information, network, geographic, and institutional frictions. The model does not require family-specific separation probabilities conditional on realized match type, family-specific preferences over nonwage amenities, or family-specific search effort.

A further feature of the final specification is persistence in wage quality across direct EE moves. In the NLSY97, 53.2 percent of direct EE moves with observed wage quality remain in the same wage tercile. An independent-draw job ladder substantially overpredicts upward wage quality mobility. I therefore allow the on-the-job search wage quality distribution to depend on the current wage-quality tercile through a common persistence parameter $\rho_E$. This parameter is common across family structure groups; the data do not provide sufficient evidence to justify a family-specific wage persistence wedge.

The model fits the eleven core moments closely. In particular, it matches pooled UE and EE transitions, the EU rate of workers from two parent households, accepted job quality by entry labor state, the strong same-quality persistence of direct EE moves, and the higher quality-standardized EU risk of workers from single parent households among E-origin matches. The estimated unemployment-search quality wedge is negative, $\widehat{\alpha}_U = -0.069$, and the on-the-job-search stability wedge is also negative, $\widehat{\gamma}_E = -0.926$. With the baseline stable opportunity share normalized to one half, the latter implies a model-generated stable share of roughly 0.284 for $SP$ workers searching while employed. The model generates a stationary unemployment rate of 6.92 percent for two parent group and 7.72 percent for the single parent group.

The mechanism decomposition shows that most of the aggregate employment instability is due to reduced access to stable opportunities during employed search. Consumption-equivalent welfare analysis suggests that employment instability of workers from single-parent households is equivalent to permanent 5.8% less consumption compared to workers from two-parent households. Existing labor policy targeting $SP$ workers, such as ASEP public-sector recruitment policy in Greece and WIOA Youth Program in Prince George's County in Maryland, are discussed as possible examples of labor policy targeting $SP$ workers to reduce employment instability of $SP$ workers.

**Related Literature** This paper relates to two broad strands of literature: (i) research on family background, family structure, and intergenerational transmission, and (ii) research on young workers' labor market dynamics and early-career instability. I also draw on search and matching models to interpret the empirical findings through the lens of accepted job quality and match durability.

First, the paper relates to research on family background, family structure, and intergenerational transmission impacts on young adulthood. A large literature studies how parental income, wealth, education, and household environment shape children's human capital accumulation and adult economic outcomes (Mayer, 1997; Bowles and Gintis, 2002; Cunha et al., 2006; Lee and Seshadri, 2019). Within this literature, studies of parental presence, household arrangements, and family disruption examine their relationship with children's educational attainment, employment, and economic mobility (McLanahan and Sandefur, 1994; Ermisch et al., 2004; Gruber, 2004; Björklund et al., 2007; McLanahan et al., 2013; Bloome, 2017; Kearney, 2023; Frimmel et al., 2024; Johnston et al., 2025). Socioeconomic impacts of family structure can also extend beyond childhood through financial support, access to the parental home, and employment networks that affect young adults' job search and labor market opportunities (Corak and Piraino, 2011; Kaplan, 2012; Kramarz and Skans, 2014; Telemo and Visschers, 2022; Holmberg et al., 2024; Staiger, 2021). Related work connects family structure and social capital to youth employment, school-to-work trajectories, and job quality (Berloffa et al., 2016; Filandri et al., 2019; Bolíbar et al., 2019; Broschinski et al., 2022). I complement this literature by studying heterogeneity in the *dynamics* of young

adult employment, after conditioning on observed human capital and financial resources.

Second, this paper relates to research on young workers' labor market dynamics and early-career instability. Young workers have high rates of mobility across jobs and labor market states, and early unemployment can have persistent consequences for later earnings and employment (Topel and Ward, 1992; Neumark, 2002; Mroz and Savage, 2006; Wachter and Bender, 2006; Kahn, 2010; Oreopoulos et al., 2012; Cajner et al., 2023). This literature establishes that the transition from school to work is a vulnerable stage of the life cycle. I add to this work by examining whether these dynamics differ systematically by family structure. This focus connects youth labor market instability to family background heterogeneity and shifts attention from job-finding rate alone and to the quality and durability of the matches young workers have.

To interpret these patterns, I estimate a quantitative job-ladder model with heterogeneous match characteristics (Sullivan and To, 2014; Baydur and Mukoyama, 2020; Jarosch, 2023). In particular, I follow the distinction in Baydur and Mukoyama (2020) between match quality and match stability, and adapt it to study heterogeneity by family structure. The structural contribution is a labor market transition model in which a predetermined family structure shifts the distribution of opportunity set relevant for job search.

Relative to these literature, this paper makes three contributions. First, it brings a flow-based perspective to the study of family structure and labor market outcomes. Rather than treating unemployment as a static outcome, the paper decomposes family-structure gaps into the transition rates that generate them. Second, it broadens the interpretation of family-background disadvantage beyond educational attainment and job-finding. The evidence points to accepted job quality and match durability as proximate margins through which early-life family structure is related to employment instability, even after adjusting for observed human capital. Third, it provides a structural decomposition of these margins. By separating accepted job quality differences from match stability differences, the model clarifies whether family-structure gaps operate mainly through the composition of jobs workers enter or through the durability of matches conditional on job quality.

The remainder of the paper proceeds as follows. Section 2 describes the NLSY97 data, the construction of family structure at age 16, and the post-school weekly labor market panel. Section 3 documents reduced-form empirical evidence on labor market flows, accepted job quality, and origin-specific employment stability. Section 4 develops the quantitative job-ladder model. Section 5 describes calibration, estimation and discusses model fit. Section 6 conducts mechanism decomposition, welfare analysis, and policy counterfactual exercises. Section 7 concludes.

# 2 Data and Measurement

## 2.1 NLSY97 and Family Structure at Age 16

I use data from the National Longitudinal Survey of Youth 1997 (NLSY97), a longitudinal survey that follows a nationally representative cohort of individuals born between 1980 and 1984 and first interviewed in 1997. Respondents were adolescents at the time of the initial interview and have been followed into adulthood. The NLSY97 is well suited for this paper for two reasons. First, it contains detailed information on respondents' family environments during adolescence, including household rosters that can be used to measure family structure around age 16. Second, it provides high frequency employment histories through event-history arrays, which allow labor market states and transitions to be measured at the weekly level.

The weekly structure of the employment histories is central to the analysis. Standard annual or point-in-time measure of employment status are useful for studying employment and unemployment levels, but they obscure the flows that generate those stocks. In contrast, the NLSY97 event-history arrays allow me to distinguish whether unemployment differences arise because some workers are less likely to find jobs, more likely to separate from jobs, or more likely to move in and out of the labor force. This distinction is particularly important for young workers, whose post-school employment histories involve frequent transitions across jobs and labor market states.

The main explanatory variable is family structure during adolescence. I construct this variable using household roster and relationship information from the NLSY97, which records the respondent's household members and their relationships to the respondent. The measure is based on household composition around at 16 and classifies respondents according to whether they lived with one resident parent or with two resident parent figures. I code respondents with one resident parent and no second resident parent figure as having a single-parent background, and respondents with two resident parent figures as having a two-parent background. Throughout the paper, I refer to these groups as $SP$ type parent and $TP$ type workers.

This definition is designed to capture the family environment during late adolescence, before most respondents enter the post-school labor market. I focus on age 16 for two reasons. First, it is late enough to capture the family structure most relevant for the transition from school to work. Second, it measures family structure before respondents' own labor market histories, residential choices, or household formation decisions can directly shape the observed family environment. This timing allows family structure to be treated as a predetermined early-life condition relative to the post-school labor market transitions studied in the paper.

## 2.2 Post-School Labor Market States and Transitions

I construct weekly labor market states from the NLSY97 employment event-history arrays. For each respondent week, I classify the respondent as employed, unemployed, or not in the labor force.

The main analysis focuses on post-school weeks. For each respondent, I use schooling histories to identify the end of the schooling spell associated with their final educational attainment, and I retain weekly employment status observations after that point. This respondent-specific restriction ensures that the analysis captures labor market dynamics after school completion, rather than employment fluctuations during periods in which respondents are still primarily students. The resulting weekly panel follows respondents through their post-school employment histories and allows me to measure movements across employment, unemployment, and nonparticipation at high frequency.

Using the weekly panel, I construct labor market transitions between consecutive weeks. The main transitions are UE transitions and EU transitions. A UE transition occurs when a respondent is unemployed in week $t$ and employed in week $t+1$. An EU transition occurs when a respondent is employed in week $t$ and unemployed in week $t+1$. Weeks involving nonparticipation are retained to measure movements into and out of the labor force and to construct supplementary transition measures, but the main empirical analysis focuses on the distinction between job-finding from unemployment and separation from employment into unemployment.

### 2.3 Wage-Based Job Quality

To study the margins underlying employment instability, I augment the weekly employment panel with two sets of measures: characteristics of accepted jobs and worker-side resources. For the main analysis, I summarize the observed quality of an accepted match using its starting hourly wage and classify jobs into low, middle, and high wage quality categories relative to the distribution of starting wages in the job entry year. This wage-based measure provides a common observable dimension of job quality that maps directly into the quantitative model. The NLSY97 also reports a broader set of job characteristics, including occupation, industry, employer size, benefits, contract arrangements, hours, and work schedules. I use these in supplementary robustness exercises reported in the Appendix [X]. The second set of measures captures worker-side resources and buffers and is described in the next subsection.

For each employed week, I identify the associated employer using the employer identifier recorded in the weekly employment history. Because job characteristics are generally reported at the employer or interview level rather than separately for every week, and because the same worker-employer match may appear in multiple interview records, I consolidate repeated records at the worker-employer-match level before linking the resulting match characteristics back to the weekly employment panel. The unit of observation in the accepted-job analysis is consequently an accepted worker-employer match rather than a respondent-week.

The main measure of observed job quality is the hourly wage at the beginning of the match. I classify each accepted job into the low, middle, or high tercile of the starting-wage distribution within the calendar year in which the match begins. Defining quality relative to the job-entry year removes changes in the nominal wage distribution over time and measures a job's position

relative to other jobs begun in the same period.

The broader NLSY97 job information provides a complementary check on this wage-based measure. These data include occupation and industry categories, employer size, union or collective-bargaining coverage, temporary-agency, on-call, and contract arrangements, hours and schedule characteristics, and employer-provided benefits. Rather than incorporating these dimensions into the main definition of job quality, I use them to test whether the family structure differences in employment stability are accounted for by other observable characteristics of accepted matches. Detailed variable construction and the corresponding robustness results are reported in Appnedix [X].

Finally, because the NLSY97 records jobs that workers accept but does not observe rejected offers, the empirical objects in this section should be interpreted as characteristics of accepted worker-employer matches, not as direct measures of the underlying firm-side offer distribution. The quantitative model therefore uses the observed composition of accepted matches to discipline latent opportunity distributions jointly with workers' acceptance decisions.

### 2.4 Family Resources and Worker-Side Buffers

Having defined the observed characteristics of accepted matches, I next turn to worker-side resources and buffers that may also be associated with post-school employment stability. The main empirical specifications control for parental education and financial resources measured before or around labor-market entry. These controls are intended to assess whether the family-structure gaps documented in the paper remain after conditioning on observable differences in the economic resources available to workers from different family backgrounds.

The first set of financial-resource measures comes from the initial NLSY97 survey round and includes parent-reported household income, net worth, and poverty status. I include missing-value indicators for these measures so that respondents with incomplete resource information remain in the estimation sample. I also construct household income and poverty measures around the respondent's school exit. The school-exit measures capture the financial environment closer to the transition into the post-school labor market and complement the earlier measures of parental resources. The main specifications sequentially add these resource blocks to the demographic, own-education, and parental-education controls.

I use a broader set of worker-side measures in supplementary robustness and mechanism exercises. These include household assets and debt observed later in young adulthood, local geographic characteristics, residential mobility around school exit, and proximity to parents. Geographic measures include metropolitan status, urban residence, and region, while parental proximity distinguishes respondents living with or near a parent from those living farther away. These variables are not part of the baseline specification; instead, they are used to assess whether the main family-structure patterns are sensitive to broader differences in workers' spatial environments and potential access to family support. The corresponding results are reported in Ap-

pendix [X].

I interpret all of these covariate adjustments descriptively. Family financial resources, residential location, and proximity to parents may themselves be related to earlier family structure and, in some cases, to subsequent labor-market choices. Accordingly, changes in the estimated family structure coefficients after adding these controls should not be interpreted as causal mediation effects. Rather, the specifications show how much of the conditional family-structure gap remains after accounting for different sets of observed resources and constraints.

### 2.5 Analysis Sample

The analysis sample consists of respondent-week observations after school exit for individuals with valid information on family structure at age 16 and nonmissing labor market states. I exclude weeks before age 16 and weeks that cannot be assigned to employment, unemployment, or nonparticipation. I also exclude observations with missing or inconsistent calendar information that prevents the construction of weekly transitions. For transition analyses, I require valid labor market states in consecutive weeks.

The main sample pools respondents across survey rounds and follows them through the available weekly employment histories. Standard errors are clustered at the respondent level to account for serial correlation in weekly observations. The empirical analysis includes controls for sex, race, birth cohort, own education, parental education, calendar time, and other baseline characteristics, depending on the specification.

## 3 Empirical Analysis

This section documents how post-school labor market outcomes differ by family structure during adolescence. I begin with weekly labor market status outcomes, distinguishing employment, unemployment, and nonparticipation. I then examine whether the unemployment gap reflects differences in transitions into employment or transitions out of employment. Next, I study the quality of accepted jobs and how accepted job quality evolves over workers' post-school employment histories. Finally, I examine employment transitions by the quality of the job at origin, distinguishing the probability that the origin worker-employer relationship ends from the probability of moving directly to another employer conditional on such an exit. This final decomposition provides the quality-specific transition moments that discipline the structural model.

### 3.1 Empirical Strategy

The empirical analysis is not designed to estimate the total effect of family structure on young adult labor market outcomes. Family structure may affect labor employment through many channels, including educational attainment and other forms of human capital accumulation. Instead,

the main empirical question is whether family structure is associated with labor market instability even after adjusting for observed human capital. I therefore include own educational attainment controls, including final grade completed and final degree category, in the main specifications. The resulting estimates should be interpreted as human capital adjusted family structure gaps rather than total family structure effects.

Throughout the empirical analysis, I use sequential covariate adjustments for two distinct purposes.[1] In the weekly status and transition hazard specifications, additional controls serve as sensitivity checks: they show whether the family-structure gaps remain after conditioning on observed human capital, parental education, and household resources. In the accepted match and worker buffer specifications, the added covariates serve as descriptive accounting controls: they show whether the higher EU hazard is attenuated after conditioning on observed match conditions and worker-side resources. I do not interpret these specifications as causal mediation estimates. Instead, they describe how the estimated family structure gaps change when conditioning on pre-specified, substantively motivated covariate blocks.

### 3.2 Post-School Weekly Labor Market Status

Table 1 first reports raw post-school weekly labor market status by family structure at age 16. Individuals from single-parent households have lower employment shares, higher unemployment, and nonparticipation shares during the post-school period. Among respondents from two-parent households, employed weeks account for 80.9% of all post-shool weeks, while unemployed weeks and weeks out of the labor force account for 4.5% unemployed and 14.6%, respectively. Among respondents from single-parent households, the corresponding shares are 71.0%, 7.5%, and 21.5%. Restricting attention to labor force weeks, the unemployment rate is 5.3% for workers from two-parent households and 9.5% for workers from single-parent households.

To assess whether these differences remain after conditioning on observed characteristics, I estimate separate binary logit models for each weekly labor market status outcome and report adjusted probability differences between workers from single-parent and two-parent households.[2] For the binary family structure indicator, the reported difference is computed by changing the indicator from two-parent to single-parent households for each observation, holding all other covariates at their observed values, and averaging the resulting change in predicted probability over the estimation sample. I therefore refer to this quantity as an *adjusted probability difference*, rather than interpreting it as the effect of a marginal increase in single parenthood.

The dependent variables indicate whether a respondent is employed, unemployed, or out of

[1] Sequential covariabe adjustments can be sensitive to the order in which correlated controls are added. Gelbach (2016) provides a formal omitted variable bias based decomposition for attributing coefficient movements to added covariates. I do not implement that decomposition here, so the sequential specifications should be interpreted descriptively rather than as an order-invariant decomposition.

[2] A multinomial logit model is a natural alternative because the three weekly states are mutually exclusive. I use separate binary logits because they allow the adjusted probability difference between the two groups to be reported directly for each labor market state in percentage point terms.

Table 1: Post-School Weekly Labor Market Status by Family Structure

| | Two-parent background | Single-parent background |
|---|---|---|
| Respondents | 5,055 | 2,583 |
| Post-school respondent-weeks | 3,851,002 | 2,080,489 |
| Employed weeks | 3,114,884 | 1,477,860 |
| Unemployed weeks | 172,727 | 155,118 |
| NILF weeks | 563,391 | 447,511 |
| Employment share | 80.89% | 71.03% |
| Unemployment share | 4.49% | 7.46% |
| NILF share | 14.63% | 21.51% |
| Unemployment rate among labor force weeks, $(U/(E+U))$ | 5.25% | 9.50% |

*Notes:* This table reports post-school respondent weeks by family structure at age 16. Respondents are counted once within each family-structure group. Shares are computed over pooled respondent-week observations within each family-structure group. The unemployment share uses all post-school weeks as the denominator, while $U/(E+U)$ restricts the denominator to labor force weeks. NILF denotes not-in- the-labor-force.

the labor force in a given post-school week. I also estimate a binary logit model for unemployment conditional on labor force participation and an individual level regression of each respondent's post-school unemployment share, $U_i/(E_i+U_i)$. The respondent week specifications control for sex, race, birth year, birth month, final grade completed, final degree category, and calender-quarter fixed effects. I then add parental education and first-round parental financial resources, including parental income, net worth, poverty status, and corresponding missingness indicators. Standard errors in the respondent-week specifications are clustered at the respondent level. The individual level unemployment share regression gives equal weight to each respondent.

Table 2 reports the most fully adjusted specifications, including demographics, own education, parental education, first-round parental resources, and calendar-quarter fixed effects. Conditional on these observed characteristics, the adjusted probability of employment is 2.0 percentage points lower for workers from single-parent households, while the adjusted probability of unemployment is 0.8 percentage points higher. Among labor force weeks, the adjusted unemployment probability is 1.0 percentage point higher for workers from single-parent households. This gap is equivalent to about 16% of the standardized two-parent benchmark of 6.1%. The NILF difference is positive but smaller and only marginally statistically significant. The individual level unemployment share regression also yields a positive and statistically significant difference, showing that the result is robust to giving each respondent equal weight rather than weighting individuals by the number of observed labor force weeks.

Taken together, these results in Table 2 point to unemployment as a key dimension of the post-school employment instability associated with family structure. Table 3 therefore focuses on unemployment among labor force weeks and examines how this gap changes as additional

Table 2: Adjusted Differences in Post-School Weekly Labor Market Status

| | All post-school weeks | | | Labor force weeks | Individual |
|---|---|---|---|---|---|
| | Employed | Unemployed | NILF | Unemployed | $U_i/(E_i+U_i)$ |
| SP–TP difference (% points) | -2.04*** | 0.76*** | 1.21* | 1.00*** | 1.44*** |
| | (0.72) | (0.23) | (0.64) | (0.30) | (0.38) |
| Standardized two-parent benchmark (%) | 79.20 | 5.18 | 15.61 | 6.15 | 7.15 |
| Observations | 5,746,202 | 5,746,149 | 5,746,202 | 4,821,227 | 7,347 |
| Respondents | 7,347 | 7,347 | 7,347 | 7,347 | 7,347 |
| Demographic controls | Yes | Yes | Yes | Yes | Yes |
| Own education controls | Yes | Yes | Yes | Yes | Yes |
| Parental education controls | Yes | Yes | Yes | Yes | Yes |
| First-round parental resources | Yes | Yes | Yes | Yes | Yes |
| Calendar-quarter fixed effects | Yes | Yes | Yes | Yes | No |

*Notes:* Columns (1)–(4)report adjusted probability differences between workers from single-parent and two-parent households from binary logit models. The reported estimate is the average discrete change in predicted probability with the single-parent household indicator is changed from zero to one, holding all other covariates at their observed values. Columns (1)–(3) use all post-school respondent weeks, while Colum (4) restricts the sample to labor force weeks. Column (5) reports the coefficient from an individual level OLS regression of each respondent's post-school unemployment share, $U_i/(E_i+U_i)$. The workers from two-parent households are the reference group. The row "Standardized two-parent benchmark" reports the average predicted outcome obtained by setting the single-parent household indicator to zero for every observation in the corresponding estimation sample while retaining all other covariates at their observed values. Standard errors are reported in parentheses and are clustered at the respondent level in Columns (1)–(4) and reported in parentheses. $***p<0.01, **p<0.05, *p<0.10$.

covariates are introduced.

In Table 3, the first specification conditions on demographics, own education, and calendar-quarter fixed effects. I then sequentially add parental education and two sets of family financial resource measures. First-round resources capture earlier household financial circumstances using parental-reported income, net worth, and poverty status from the initial NLSY97 survey, whereas school-exit resources measure household income and poverty closer to the respondent's transition out of schooling. The latter provide a more proximate measure of family financial circumstances around labor market entry and allow us to assess whether the unemployment difference is sensitive to changes in household resources between the initial survey and school exit period.

The adjusted unemployment difference is 1.35 percentage points in the baseline specification and changes only slightly, to 1.28 percentage points, after adding parental education. Adding first-round parental resources reduces the difference to 1.00 percentage point. When resources measured around school exit are also included, the difference falls further to 0.61 percentage points and remains statistically significant at the 5 percent level. Thus, the estimated difference becomes smaller as observed family financial resources are added, particularly when conditioning on resources measured closer to labor market entry. Because these resources may themselves be related to earlier family structure and subsequent labor market choices, I interpret the sequential changes descriptively rather than as causal mediation.

One remaining concern is that the association between family structure and unemployment

Table 3: Labor Force Unemployment Gap and Family-Background Controls

| | Probability of Unemployment among labor force weeks | | | |
|---|---|---|---|---|
| | Baseline | + Parent edu. | + R1 resources | + Exit resources |
| SP–TP difference (% points) | 1.35*** | 1.28*** | 1.00*** | 0.61** |
| | (0.30) | (0.30) | (0.30) | (0.29) |
| Standardized two-parent benchmark (%) | 6.08 | 6.10 | 6.15 | 6.31 |
| Observations | 4,824,527 | 4,824,527 | 4,821,227 | 4,821,227 |
| Respondents | 7,406 | 7,406 | 7,347 | 7,347 |
| Demographic controls | Yes | Yes | Yes | Yes |
| Calendar-quarter fixed effects | Yes | Yes | Yes | Yes |
| Birth year/month controls | Yes | Yes | Yes | Yes |
| Own education controls | Yes | Yes | Yes | Yes |
| Parental education controls | No | Yes | Yes | Yes |
| First-round parental resources | No | No | Yes | Yes |
| School-exit resources | No | No | No | Yes |

*Notes:* The outcome is an indicator for unemployment among labor force respondent weeks. The table reports adjusted probability differences between workers from single-parent and two-parent households from logit models. For each specification, the reported estimate is the average discrete change in predicted unemployment probability when the single-parent household indicator is changed from zero to one, holding all other covariates at their observed values. The "Standardized two-parent benchmark" is the average predicted unemployment probability obtained by setting the single-parent household indicator to zero for all observations. Standard errors are clustered at the respondent level and reported in parentheses. . $***p < 0.01, **p < 0.05, *p < 0.10$.

that remains after conditioning on education and financial resources may reflect persistent worker characteristics not captured by these controls, including noncognitive traits. This possibility is motivated by the skill-formation literature, which emphasizes that family environments can shape persistent cognitive and noncognitive abilities that matter for later socioeconomic outcomes (Cunha et al., 2006). Appendix A examines this possibility using the noncognitive measures available in the NLSY97: Goldberg-based measures of conscientiousness and agreeableness, together with an earlier measure of behavioral and emotional problems.

These observed noncognitive characteristics are predictive of unemployment, with higher conscientiousness in particular associated with a lower unemployment probability. However, adding these measures reduces the estimated association between family structure and unemployment by only about 9–16 percent in same-sample comparisons. This relatively small amount of attenuation suggests that noncognitive characteristics may account for part of the remaining association, but it is not the primary underlying mechanism behind the higher unemployment rate of workers from single-parent households. Appendix A describes the detailed results and discusses the limitations of this interpretation.

### 3.3 UE and EU Transition Hazards

The weekly status results show that workers from single-parent households have higher unemployment rates, but unemployment is a stock variable. A higher unemployment rate can arise because workers enter into unemployment more often, because they exit from unemployment more slowly, or both. I therefore decompose the unemployment gap into weekly transition hazards.

I estimate discrete-time transition hazard models for two transitions by using a complementary log-log specification.[3] The first transition is the unemployment-to-employment transition, UE, where the risk set consists of unemployed respondent weeks and the outcome equals 1 if the worker is employed in the following week. The second transition is the employment-to-unemployment transition, EU, where the risk set consists of employed respondent weeks and the outcome equals 1 if the worker is unemployed in the following week. The empirical specification is

$$\log[-\log(1-h_{it})] = \alpha + \beta \text{SingleParent}_i + X_i'\gamma + \lambda_t + \delta \log D_{it}$$

where $h_{it}$ is the conditional probability that respondent $i$ makes the relevant transition in week $t$, given that the respondent is in the corresponding risk set at the beginning of the week. $\text{SingleParent}_i$ is an indicator for living with a single parent at age 16, $X_i$ includes demographic and education controls, $\lambda_t$ denotes calendar-time controls, and $D_{it}$ is the number of weeks elapsed in the current spell. In extended specifications, I add parental education and household resource controls. Standard errors are clustered at the respondent level. I report exponential coefficient, $\exp(\beta)$, as hazard ratios.

Table 4 shows that the unemployment gap is driven primarily by higher transitions from employment into unemployment rather than by slower transitions from unemployment into employment. The estimated UE hazard ratio between $SP$ workers to $TP$ workers is close to 1 and statistically insignificant across specifications. This result implies that UE risk of workers is not basically different across two family-structure groups. In contrast, the EU hazard ratio is 1.148 in the main specification, which implies that workers from single-parent households face roughly a 15 percent higher weekly risk of moving from employment to unemployment. Adding first-round parental resources reduces the hazard ratio to 1.126, and adding both first-round and school-exit resources reduces it further to 1.105. The estimate remains above one and statistically significant throughout.

Put together, these transition results indicate that the higher unemployment associated with single-parent household is not primarily explained by a slower reemployment transition once

[3]Note that labor market status data is in weekly frequency, rather than exact dates. All spell durations are grouped into weekly discrete time periods. For an alternative specification, I can estimate continuous-time Cox hazard models. However, using continuous-time models requires additional assumptions, such as when transitions take place within the week. Therefore, I use a complementary log-log link, which preserves the proportional hazards interpretation that applies to the continuous-time model without any additional unnecessary assumptions.

Table 4: Weekly Transition Hazards by Family Structure

| | UE Transition | | | EU Transition | | |
|---|---|---|---|---|---|---|
| | Main | + R1 resources | + R1 and exit resources | Main | + R1 resources | + R1 and exit resources |
| *SP*-to-*TP* hazard ratio | 0.9647 | 0.9861 | 1.0072 | 1.1481*** | 1.1262*** | 1.1048*** |
| | (0.0339) | (0.0361) | (0.0369) | (0.0400) | (0.0395) | (0.0398) |
| Observations | 325,146 | 321,635 | 321,635 | 4,539,960 | 4,539,960 | 4,539,960 |
| Transition events | 7,533 | 7,533 | 7,533 | 8,842 | 8,842 | 8,842 |
| Demographic controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Own education controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Parental education controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Calendar-time controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Duration controls | Yes | Yes | Yes | Yes | Yes | Yes |
| First-round parental resources | No | Yes | Yes | No | Yes | Yes |
| School-exit resources | No | No | Yes | No | No | Yes |

*Notes:* This table reports exponentiated coefficients from discrete-time complementary log-log transition-hazard models. Columns (1)–(3) estimate UE transitions on unemployed respondent-weeks. Columns (4)–(6) estimate EU transitions on employed respondent weeks. The coefficient on single-parent household is reported as a hazard ratio. A hazard ratio above one indicates a higher weekly transition hazard relative to workers from two-parent households. Standard errors are clustered at the respondent level and reported in parentheses. Demographic controls include sex, race, and birth cohort. Own education controls include completed schooling measures. First-round parental resources include parental income, net worth, poverty status, and missingness indicators. School-exit resources include income and poverty measures around school exit and missingness indicators. $***p < 0.01, **p < 0.05, *p < 0.10$.

unemployment has begun. Instead, the main difference appears on the employment side of the labor market: workers form single-parent households are more likely to move from employment into unemployment.

The fact that the EU difference remains after conditioning on observed human capital and family resources raises the next question: does the employment instability associated with single-parent household reflect sorting into different types of accepted jobs, or greater fragility even among jobs of similar observed quality? I examine these margins in two steps. I first study the wage-quality composition of accepted jobs. I then distinguish jobs by whether they are entered from unemployment or from employment, since job search behavior and the opportunities available to workers can differ substantially across these two labor market states (Holzer, 1987; Blau and Robins, 1990; Faberman et al., 2022). I then examine whether family-structure differences in subsequent separation risk are still present after conditioning on observed job quality, separately for jobs entered from unemployment and from employment.

An alternative explanation is that employment stability depends partly on parental proximity and informal family support. An additional family member may supply time or logistical assistance that partly offsets differences in the worker-side buffers associated with single-parent household. Appendix B examines this possibility using grandparent availability and the result provides a *suggestive* evidence of time-transfer channel underlying higher EU transition rates of *SP* workers. Because this analysis does not directly observe actual transfers of time, transportation, care, or other forms of assistance, and may themselves reflect earlier family circumstances and endogenous location or co-residence choices, I interpret the results as descriptive mechanism

diagnostic rather than as causal estimates.

## 3.4 Accepted Job Quality

### 3.4.1 Average Differences in Accepted Job Quality

Having established that the unemployment gap by family structure is driven primarily by a higher EU hazard rather than by a lower job-finding rate, I next examine whether workers from different family structure enter systematically different types of accepted jobs. I use the hourly wage at the beginning of each worker-employer match as an observable proxy for accepted-job quality.[4] Specifically, I classify each accepted job into the low, middle, or high tercile of hourly wage distribution. Thus, throughout this subsection, job quality refers to a wage-based measure of the quality of accepted jobs.

I estimate multinomial logit models of accepted-job quality and report average discrete change associated with single-parent household in the probability of entering the low and high wage terciles. The specifications sequentially add job-entry-year and accepted-job-order fixed effects, demographic characteristics, parental education, and the worker's own educational attainment.

Table 5 reports the results. In the unadjusted specification, workers from single-parent hoursholds are 9.1 percentage points more likely to enter the low wage tercile and 10.5 percentage points less likely to enter the high wage tercile. These differences change littel after adding job-entry-year and accepted-job-order fixed effects, but become progressively smaller as demographic characteristics, parental education, and the worker's own education are added.

In the full specification, single-parent household is associated with a 2.1 percentage point higher probability of entering the low wage tercile and a 1.9 percentage point lower probability of entering the high wage tercile. The former estimate is statistically significant at the 5 percent level, while the latter is significant at the 10 percent level. Thus, much of the raw difference in accepted-job quality is associated with observed worker characteristics and family resources, but a modest difference in the composition of accepted jobs remains after conditioning on these observables.

### 3.4.2 Accepted Job Quality over Employment Histories

The previous pooled estimates show that workers from single-parent households enter lower quality jobs on average. I next examine whether this job-quality disadvantage is confined to initial labor market entry or recurs as workers move through subsequent jobs. A disadvantage that reappears after the first job would indicate that the accepted-job quality gap is a recurring feature of the employment history, rather than a one-time condition at school exit. I therefore compare accepted-job quality at the first and subsequent post-school jobs and then distinguish subsequent jobs by the worker's labor market state immediately before job entry.

[4]The wage-based measure captures one observable dimension of job quality and should not be interpreted as a comprehensive measure of nonwage amenities, working conditions, or job security.

Table 5: Family Structure and Accepted Job Quality

| | *SP–TP* predicted probability difference | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Low-quality job (% points) | 9.08*** | 8.91*** | 6.05*** | 4.60*** | 2.11** |
| | (1.03) | (1.03) | (1.03) | (1.04) | (0.98) |
| High-quality job (% points) | −10.48*** | −10.31*** | −7.08*** | −4.69*** | −1.87* |
| | (1.07) | (1.07) | (1.10) | (1.09) | (1.05) |
| Entry-year fixed effects | No | Yes | Yes | Yes | Yes |
| Accepted-job-order fixed effects | No | Yes | Yes | Yes | Yes |
| Demographic controls | No | No | Yes | Yes | Yes |
| Parental-education controls | No | No | No | Yes | Yes |
| Own-education controls | No | No | No | No | Yes |
| Accepted-job observations | 34,482 | 34,482 | 34,482 | 34,482 | 34,113 |

*Notes:* The table reports average marginal effects from multinomial logit models of accepted-job quality. Job quality is measured using terciles of start-year-adjusted hourly wages, with the middle tercile serving as the base outcome. The reported coefficients are the average discrete change in percentage points, in the probability of entering the indicated quality category associated with single parent household, relative to two parent household. The unit of observation is an accepted worker-employer match.

Accepted-job-order fixed effects distinguish job orders one through ten and pool orders eleven and above. Demographic controls include sex, race, and birth-year indicators. Parental-education controls include categorical measures of maternal and paternal educational attainment. Own-education controls include final grade completed and final educational-attainment category. Standard errors, clustered at the worker level, are reported in parentheses.

The estimates represent conditional associations. The attenuation across specification should not be interpreted as a causal mediation decomposition. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6 examines where the accepted job quality gap arises over post-school employment histories. Panel A shows that the accepted-job quality disadvantage is not confined to the first post-school job. At the first accepted job, single-parent household is associated with a 4.82 percentage point lower probability of entering the high wage tercile, while the corresponding difference in the low wage tercile is positive but statistically insignificant. Among subsequent jobs, the pattern shifts toward the lower tail of the wage distribution: single-parent household is associated with a 2.17 percentage point higher probability of entering the low wage tercile, while the difference in the high wage tercile is smaller and statistically insignificant. Thus, the disadvantage reappears after the initial school-to-work transition, although the margin along which it appears changes changes over the employment history.

Panel B then distinguishes subsequent accepted jobs by the worker's labor market state immediately before job entry. The largest differences appear among jobs entered from unemployment. Conditional on the full set of controls, workers from single-parent households are 5.70 percentage points more likely to enter the low wage tercile following unemployment and 4.25 percentage points less likely to enter the high wage tercile. By contrast, there is little evidence of a comparable difference among jobs entered directly from employment: the estimated differences in both the low and high wage terciles are close to zero. The estimates for jobs entered from nonparticipation

Table 6: Accepted Job Quality over Post-School Employment Histories

| | *SP–TP* predicted probability difference | | Sample size | |
|---|---|---|---|---|
| | Low tercile (% points) | High tercile (% points) | Jobs | Workers |
| ***Panel A. First and subsequent accepted jobs*** | | | | |
| First post-school accepted job | 1.32 | $-4.82^{***}$ | 3,965 | 3,965 |
| | (1.44) | (1.70) | | |
| Subsequent accepted jobs | $2.17^{**}$ | $-1.48$ | 30,148 | 5,660 |
| | (1.06) | (1.11) | | |
| ***Panel B. Subsequent accepted jobs by search state*** | | | | |
| Employment | 0.72 | 0.07 | 10,920 | 3,968 |
| | (1.31) | (1.58) | | |
| Unemployment | $5.70^{***}$ | $-4.25^{**}$ | 4,825 | 2,465 |
| | (1.87) | (1.78) | | |
| Not-in-the-labor-force | 1.88 | $-1.06$ | 13,661 | 4,470 |
| | (1.32) | (1.24) | | |

*Notes:* The table reports average marginal effects, in percentage points, of a single parent household on the probability that an accepted job falls in the low or high tercile of the starting-wage distribution. The middle wage tercile is the omitted outcome. Wage terciles are defined relative to the calendar year in which the job begins. The unit of observation is an accepted worker-employer match.

Panel A estimates separate multinomial logit models for the first post-school accepted job and for subsequent accepted jobs.

Panel B estimates separate multinomial logit models according to the worker's labor market state immediately before entry into the subsequent accepted job. Employment, unemployment, and not-in-the-labor-force denote the three reported origin states. Subsequent jobs with another or unclassified origin state are not reported in Panel B.

All specifications include job-entry-year fixed effects, sex, birth-year fixed effects, race, maternal and paternal education, final grade completed, and final educational-attainment category. Standard errors are clustered by worker and reported in parentheses. Workers with multiple subsequent accepted jobs contribute multiple observations. All accepted jobs begin after the respondent's terminal observed schooling period.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

are also smaller and statistically insignificant.

This result sharpens the interpretation of the transition results in Table 4. Workers from single-parent households do not have substantially lower UE transition rates, but conditional on reemployment, the *composition* of accepted jobs is less favorable. The difference is therefore not primarily in whether workers leave unemployment, but in the types of jobs they enter when they do.

These results establish that the accepted-job quality disadvantage is recurring and is especially pronounced for jobs entered from unemployed. They do not, however, show whether workers from different family structure subsequently experience different employment stability *within* jobs of similar observed quality. I examine this margin in the next subsection.

### 3.5 Separating Job-Quality Composition from Match Stability

The previous subsection establishes a family-structure difference in the *composition* of accepted jobs: workers from single-parent households are more likely to enter low quality jobs, particularly following unemployment. Because lower wage jobs tend to be associated with higher separation risk (Jarosch, 2023), this composition difference could itself contribute to the higher EU hazard documented in Table 4. The relevant question is therefore whether the family-structure difference in subsequent employment stability remains once matches are evaluated under a common distribution of observed job quality.

I next examine whether there is an additional stability channel beyond this sorting margin. To separate job-quality composition from subsequent match stability, I classify each worker-employer match by its entry path. I distinguish jobs reached through a direct job-to-job transition from those entered from unemployment. I then estimate a weekly complementary log-log model of the subsequent EU transition that fully interacts family structure, entry path, and the wage quality tercile of the accepted match. The specification includes demographic characteristics, own and parental education, household financial resources, calendar time, and employment duration.

I use estimated model to obtain adjusted weekly EU hazards for each family-structure×entry-path×wage-tercile cell. Within each entry path, I then aggregate the three tercile-specific predictions using a *common* distribution of observed job quality. The common weights are calculated from the pooled distribution of accepted matches across two family-structure groups, assigning one observation to each distinct worker-employer match. The same low, middle, high wage shares are therefore applied to workers from single-parent and two-parent households among jobs reached through job-to-job transitions and, separately, among jobs entered from unemployment. This standardization removes family-structure differences in the wage quality composition of accepted jobs without requiring the two entry paths to have the same quality distribution.

Table 7 reports the resulting $SP$-to-$TP$ hazard ratios. Hazard ratios between two family-structure groups represents how much more likely $SP$ workers are to experience that event. Among jobs reached through direct job-to-job transitions, the job-quality-standardized EU hazard is 14.5 percent higher for workers from single-parent households. The estimate is statistically significant at the 5 percent level ($p = 0.038$). Thus, equalizing the distribution of observed job quality does not eliminate the family-structure difference in subsequent employment stability among matches reached through employed search.

The corresponding pattern is not observed among jobs entered from unemployment. The standardized $SP$-to-$TP$ hazard ratio is 0.979 and is not statistically distinguishable from 1 ($p = 0.759$). A direct test of equality between the two entry-path-specific hazard ratios yields a $p$-value of 0.053. This provides a suggestive evidence that the family-structure stability gap is larger among matches reached through job-to-job transitions.

The results therefore indicate different roles for observed job characteristics across entry paths. Among jobs entered from unemployment, the main family-structure difference lies in the quality

Table 7: Job-Quality-Standardized EU Hazard Ratios by Match Entry Origin

| | E-origin matches<br>Job-to-job entry | U-origin matches<br>Entry from unemployment |
|---|---|---|
| $SP$-to-$TP$ hazard ratio | 1.1454** | 0.9794 |
| | (0.0751) | (0.0665) |
| Accepted matches | 11,133 | 4,950 |
| Full controls | Yes | Yes |
| Joint risk-week observations | 1,650,920 | |
| EU transitions | 2,960 | |
| Workers | 4,923 | |

Notes: The table reports $SP$-to-$TP$ hazard ratios from a discrete-time complementary log-log model of weekly employment-to-unemployment transitions. Cell-specific adjusted hazards are aggregated using the pooled match-level distribution of wage terciles within each entry origin, with the same origin-specific weights applied to both family-structure groups. The specification includes demographic characteristics, own and parental education, parental and school-exit resources, calendar time, and employment duration. Delta-method standard errors of the reported hazard ratios are shown in parentheses and are clustered by worker. Significance levels are based on tests of $H_0 : HR = 1$. The $p$-value for equality of the two origin-specific hazard ratios is 0.053. ** $p < 0.05$.

distribution documented in Table 6. Once that quality distribution is equalized, there is no evidence of an additional EU hazard gap in jobs reached from unemployment. Among jobs reached through direct job-to-job transitions, by contrast, equalizing the distribution of observed job quality leaves a positive and statistically significant stability gap. This residual difference points to a dimension of match durability not summarized by the wage tercile of the accepted job.

### 3.6 Summary of Empirical Findings

The empirical analysis establishes three main facts. First, workers from single parent households experience higher post-school unemployment rate even after conditioning on observed human capital, parental education, and household resources. The difference is driven primarily by a higher employment-to-unemployment hazard rather than by a lower unemployment-to-employment hazard.

Second, workers from single-parent households enter a less favorable distribution of accepted jobs. This difference is not confined to the initial school-to-work transition and is particularly pronounced among jobs obtained following unemployment. Thus, similar rates of reemployment do not imply similar quality of reemployment.

Third, the family difference in subsequent employment stability varies with the path through which a job is entered. After standardizing for observed job quality distribution, the $SP$-to-$TP$ EU hazard ratio is greater than 1 among jobs reached through direct job-to-job transitions, whereas the hazard ratio is close to one among jobs entered from unemployment.

These facts motivate a distinction between the *quality distribution* of job offer accessible from

unemployment and the *stability distribution* of job offer accessible during employed search. The quantitative model formalizes this distinction by allowing family structure to shift the wage-quality distribution of opportunities accessible from unemployment and the stability composition of opportunities accessible while employed. These opportunity distributions, together with workers' acceptance decisions and job-ladder mobility, generate the patterns of accepted job and employment transition observed in the data.

# 4 Quantitative Job-Ladder Model

In this section, I develop a parsimonious job-ladder model to quantify how differences in job opportunity distribution can generate the family structure gap documented in Section 3. I build on the job-ladder framework of Baydur and Mukoyama (2020), extending their model of heterogeneous match characteristics to allow the distribution of quality and stability composition to differ by family structure. The model is not a model of family behavior or parental inputs. Family structure enters only as a predetermined group characteristic indexing a reduced-form difference in the distribution of job-offer. Conditional on the characteristics of a realized match, workers from the two family-background groups face the same separation technology.

Note that the model focuses on employment and unemployment and abstracts from labor-force nonparticipation. Workers may move from unemployment into employment, remain with their current employer, move directly to another employer, or separate into unemployment. Accordingly, the quantitative moments are constructed from worker-employer matches whose observed entry origins lie in employment or unemployment.[5]

## 4.1 Environment

Time is discrete and measured in weeks. Workers belong to family structure group

$$g \in \{TP, SP\},$$

where $TP$ denotes a two-parent household and $SP$ denotes a single-parent household at age 16.

A worker is either unemployed, $U$, or employed, $E$ in a match

$$m = (q, x, s).$$

The first match characteristic is observed wage quality,

$$q \in \{L, M, H\}.$$

[5]The full sample EU hazard estimates in Table 4 remain important as reduced-form facts, but they are not mechanically identical to model moments because the empirical sample also contains jobs entered from nonparticipation.

It corresponds to the low, middle, and high terciles of the starting-wage distribution. Let $w_q$ denote the normalized flow payoff associated with quality $q$.

The second component, $x$, captures nonwage match value. I assume

$$x \sim N(0, \sigma_x^2).$$

The mean-zero assumption centers $x$ around the average match, so that $x$ captures how the nonwage value of a particular match differs from that of an average match. Workers observe the realization of $x$ when evaluating a job offer.

The third component,

$$s \in \{F, S\}$$

is latent match stability. Fragile matches separate into unemployment with weekly probability $\delta_F$, while stable matches separate with probability $\delta_S$, where

$$\delta_F > \delta_S.$$

Note that $\delta_F$ and $\delta_S$ are common across family structure groups. Family structure therefore does not directly alter the separation technology conditional on realized match type.

The timing at the offer stage follows Baydur and Mukoyama (2020). When a job offer arrives, the worker observes *all three* match attributes $(q, x, s)$ before accepting or rejecting it.

An unemployed worker receives a job offer with probability $\lambda_U$ and an employed worker receives with $\lambda_E$. I assume no voluntary move from employment to unemployment. For an employed worker in match $m = (q, x, s)$, events occur sequentially within the week. The current match first separates with probability $\delta_s$. Conditional on survival, a job offer arrives with probability $\lambda_E$. If an offer arrives, the worker observes all of its characteristics $(q', x', s')$ and accepts it only if the prospective match yields a higher continuation value than the current match.

### 4.2 Distributions of Match Characteristics

Let $m' = (q', x', s')$ be a newly arriving job offer. Note that the prime notation distinguishes the prospective match from a worker's current match $m = (q, x, s)$.

Conditional on an offer arriving with probability $\lambda_o$ where current labor market state $o \in \{U, E\}$, $m'$ follows joint probability distribution $G_o^g = P_{q,o}^g \otimes F_x \otimes \Pi_{s,o}^g$. Here, $P_{q,o}^g, F_x, \Pi_{s,o}^g$ represent probability measure governing wage quality $q$, nonwage match value $x$, and stability $s$, respectively.

The empirical restrictions are imposed through the following current labor state-specific distributions. Note that results from Table 6 guide us to let the wage-quality wedge of single-parent household appear only in U-origin jobs and results from Table 7 guide us to let the stability wedge of single-parent household appear only in E-origin jobs.

To be more specific, for offers received while unemployed,

$$dG_U^g(q', x', s') = p_U^g(q'; \alpha_U) f_x(x') \pi_U(s') dx'.$$

Family structure can therefore affect the wage quality composition of these offers through the parameter $\alpha_U$, while the distributions of nonwage value and match stability are common across family types.

For offers received while employed,

$$dG_E^g(q', x', s' \mid q) = p_E(q' \mid q) f_x(x') \pi_E^g(s'; \gamma_E) dx'.$$

The transition kernel $p_E(q' \mid q)$ allows prospective quality to depend on current quality and captures persistence along the job ladder. It is common across family types. Family structure instead affects the stability composition of offers through $\gamma_E$. These distributions describe the composition of prospective matches conditional on an offer arriving. They are distinct from the offer-arrival probabilities $\lambda_E$ and $\lambda_U$, which govern how frequently workers receive offers.

Section 5 specifies the functional form used for $p_U^g(q'; \alpha_U)$, $p_E(q' \mid q)$, and $\pi_E^g(s'; \gamma_E)$ and explains how the two family-specific margins are disciplined by the data.

### 4.3 Worker Values and Acceptance Decisions

Let $V_U^g$ denote the value of unemployment for a worker from family structure group $g$, and let $V_E^g(m)$ denote the value of employment in match $m = (q, x, s)$. The flow payoff from unemployment is $\log b$, while the flow payoff from employment is

$$y(q, x) = \log w_q + x.$$

**Unemployed Workers** An unemployed worker receives a job offer with probability $\lambda_U$. Because the worker observes the prospective match before deciding, an offer $m'$ is accepted if and only if

$$V_E^g(m') \geq V_U^g.$$

The unemployment value therefore satisfies

$$V_U^g = \log b + \beta \left[ (1 - \lambda_U) V_U^g + \lambda_U \mathbb{E}_{m' \sim G_U^g} \max\{V_U^g, V_E^g(m')\} \right].$$

Define the probability of accepting an offer as

$$A_U^g = \Pr_{m' \sim G_U^g} [V_E^g(m') \geq V_U^g].$$

A UE transition thus requires both arrival of an offer and endogenous acceptance of that opportunity.

**Employed Workers** For an employed worker, events occur sequentially within the week. The current match first separates with probability with probability $\delta_s$. Conditional on survival, an offer arrives with probability $\lambda_E$. The worker observes the prospective match and moves to the new employer only if it yields a higher value than the current match. The employment value is

$$V_E^g(m) = \log w_q + x + \beta \left[ \delta_s V_U^g + (1-\delta_s) \left\{ (1-\lambda_E) V_E^g(m) + \lambda_E \mathbb{E}_{m' \sim G_E^g(\cdot|q)} \max\{V_E^g(m), V_E^g(m')\} \right\} \right].$$

Define the probability of accepting the new offer as

$$A_E^g(m) = \Pr_{m' \sim G_E^g(\cdot|q)} \left[ V_E^g(m') \geq V_E^g(m) \right].$$

Then the probability of a direct EE move from current match $m$ is

$$(1-\delta_s)\lambda_E A_E^g(m),$$

whereas the probability of remaining with the current employer is

$$(1-\delta_s)\left[1 - \lambda_E A_E^g(m)\right].$$

### 4.4 Labor Market Dynamics

The model distinguishes the primitive events that generate observed labor market transitions. The acceptance probabilities derived in the previous subsection imply the following weekly transition hazards:

$$h_{UE}^g = \lambda_U A_U^g, \qquad h_{EU}(m) = \delta_s, \qquad h_{EE}^g(m) = (1-\delta_s)\lambda_E A_E^g(m).$$

For a worker in match $m$, an EU transition $h_{EU}(m)$ occurs with probability $\delta_s$. An EE transition $h_{EE}^g(m)$ requires the current match to survive, an offer to arrive, and that offer to be accepted. A UE transition $h_{UE}^g$ similarly requires both offer arrival and acceptance. This distinction allows observed transition rates to differ even when workers face common primitive arrival or separation parameters.

The distribution of workers across unemployment and heterogeneous matches evolves according to these transition probabilities. Given optimal acceptance rules, the model generates a group-specific stationary distribution over unemployment and employment states.

In the quantitative implementation, unemployment constitutes one state, while each discretized

match $m = (q, x, s)$ constitutes a separate employment state. Let $K^g$ denote the resulting Markov transition matrix for group $g$. Each row represents a worker's current state, and each column represents a possible state in the following week. The entries of $K^g$ give the corresponding one-week transition probabilities, and each row sums to one.

Let $\mu_t^g$ denote the cross-sectional distribution of group $g$ workers across these states at the beginning of week $t$, and let $\mu_0^g$ denote the initial distribution. The component $\mu^g(U)$ is the share of the group that is unemployed, while $\mu_t^g(m)$ is the share employed in match state $m$. Let $\mu^g$ denote the stationary distribution to which this cross-sectional distribution converges. Accordingly, $\mu^g(U)$ is the group's stationary unemployment share, while $\mu^g(m)$ is its stationary share in match state $m$.

The cross-sectional distribution evolves according to

$$\mu_{t+1}^g = \mu_t^g K^g,$$

which implies

$$\mu_t^g = \mu_0^g (K^g)^t.$$

Under the standard ergodicity condition, the stationary distribution can be characterized as the long-run limit

$$\mu^g = \lim_{t \to \infty} \mu_0^g (K^g)^t.$$

Equivalently, it is the normalized fixed point satisfying

$$\mu^g = \mu^g K^g, \qquad \mu^g(U) + \sum_m \mu^g(m) = 1.$$

Although the offer arriving probabilities and separation probabilities are common across family structure groups, their transition matrices and stationary distributions need not be. Differences in prospective match distributions affect which offers workers accept and which types of matches they accumulate along the job ladder.

This distinction is central to the models' explanation of the aggregate EU gap. The group-level EU hazard is an average of $\delta_F$ and $\delta_S$, weighted by the stationary shares of employed workers in fragile and stable match. It can therefore differ across groups even though the separation probability associated with each stability type is identical.

The two family-specific wedges affect the composition of job offers through distinct margins. The quantitative implementation in Section 5 determines how these margins and remaining model parameters are parameterized and disciplined by the data.

# 5 Quantitative Implementation of the Model

## 5.1 Parameterization and Calibration

I now specify the functional forms and numerical implementation used to take the model to the data.

**Payoffs and wage quality** Note that the model abstracts from educational attainment and financial resources. In order to remove the component related to these and other observables in observed starting wages in the NLSY97, I regress log starting wages on the same set of observables used as controls in Table 5–7 to get residualized wage quality. I then divide the residualized wage distribution into terciles, which define the model's three wage-quality states.

$$\log w = X\beta + \varepsilon,$$
$$q = \text{tercile of } \varepsilon.$$

I then normalize the middle wage node to 1 by using the residualized start-year wage terciles.

The relative wage payoff nodes are

$$\mathrm{w}_q = (w_L, w_M, w_H) = (0.61157,\ 1,\ 2.12938).$$

This normalization preserves the empirical low-to-middle and high-to-middle wage ratios, but it does not yet place employment and unemployment resources on a common scale. To establish that scale, I normalize unemployment resources, including the UI-equivalent component, to $b = 1$ and calibrate the unemployment-to-middle wage ratio as

$$\omega = \frac{b}{w_M} = 0.425.$$

This scale is based on the U.S. average of the Department of Labor's claimant-level UI Replacement Ratio 1.[6] These two conditions imply

$$\tilde{w}_M = \frac{b}{\omega} = 2.35294.$$

I then preserve the empirical relative wage schedule when constructing the remaining employment-resource nodes:

$$\tilde{\mathrm{w}}_q = \frac{\mathrm{w}_q}{\omega} = (1.43900,\ 2.35294,\ 5.01031).$$

[6]The Department of Labor defines Replacement Ratio 1 as the weighted average of the weekly benefit amount divided by the claimant's normal hourly wage times 40 hours.

**Distribution of wage-quality** To parameterize ordered shifts in wage quality, define

$$Q_L = -1, \qquad Q_M = 0, \qquad Q_H = 1.$$

The index $Q_q$ is not an additional observed job characteristic. It is a coding device that allows the model to parsimoniously represent a shift toward higher or lower wage quality states and distance between ordered states.

The baseline wage quality probabilities take the multinomial logit form:

$$p_o(q) = \frac{\exp(\eta_{o,q})}{\sum_{k\in\{L,M,H\}} \exp(\eta_{o,k)}}, \qquad o \in \{U, E\},\ \eta_{o,M} = 0.$$

Note that I allow the baseline wage quality distribution of arriving offers to differ by the worker's search state. The normalization $\eta_{o,M} = 0$ makes $\eta_{o,L}$ and $\eta_{o,H}$ the log odds of low and high quality relative to middle quality.

For offers received while unemployed, $TP$ workers draw from the baseline distribution. The corresponding distribution for $SP$ workers is

$$p_U^{SP}(q) = \frac{\exp(\eta_{U,q} + \alpha_U Q_q)}{\sum_{k\in\{L,M,H\}} \exp(\eta_{U,k} + \alpha_U Q_k)}.$$

A negative $\alpha_U$ shifts the wage-quality composition of opportunities accessible to $SP$ workers toward lower-quality jobs. The parameter changes the composition of arriving offers, not their arrival frequency.

For offers received while employed, the wage-quality distribution also depends on the worker's current wage-quality state:

$$p_E(q' \mid q) = \frac{\exp(\{\eta_{E,q'} - \rho_E|Q_{q'} - Q_q|\})}{\sum_{k\in\{L,M,H\}} \exp(\{\eta_{E,k} - \rho_E|Q_k - Q_q|\})}$$

A positive $\rho_E$ makes nearby wage quality states relatively more likely and therefore captures persistence along the job ladder. This kernel is common across family structure groups.

**Distribution of nonwage match value** Nonwage match value is assumed to follow normal distribution:

$$x \sim N(0, \sigma_x^2).$$

Its mean is normalized to zero, and the distribution is represented on a 51-point grid in the benchmark computation. I calibrate the dispersion $\sigma_x$ using the fraction of direct EE wage quality changes that are upward,

$$Pr(q' > q \mid E, q' \neq q).$$

This moment identifies the strength of observed wage quality relative to unobserved nonwage value in workers' acceptance decision. When $\sigma_x$ is small, accepted wage-quality changes are predominantly upward. As $\sigma_x$ rises, a sufficiently favorable nonwage draw can compensate for a lower observed wage node.

The empirical conditional upward share is 0.61031. The calibration gives

$$\widehat{\sigma}_x = 0.99374.$$

I condition on $q' \neq q$ because the same-quality share of direct job-to-job transitions is used separately to discipline $\rho_E$, which is a quality persistence parameter.

**Distribution of match stability** I set the baseline stable-offer share to

$$\nu_0 = 0.5.$$

This baseline applies to offers received by $TP$ workers while employed and to offers received by while unemployed by both family groups. For $SP$ workers searching on the job, the stable share is

$$\nu^{SP} = \Pr(S \mid E, SP) = \Lambda\left[\text{logit}(\nu_0) + \gamma_E\right],$$

where

$$\Lambda(z) = \frac{1}{1+\exp(-z)} \qquad \text{and} \qquad \text{logit}(p) = \log\frac{p}{1-p}.$$

Equivalently,

$$\log\frac{\nu^{SP}}{1-\nu^{SP}} = \log\frac{\nu_0}{1-\nu_0} + \gamma_E.$$

Thus $\gamma_E$ is the $SP$ shift in the log odds of accessing a stable rather than a fragile employed-search opportunity relative to the $TP$ baseline. Because $\nu_0 = 0.5$, the baseline log odds are zero and the $SP$ stable share simplifies to $\Lambda(\gamma_E)$. A negative $\gamma_E$ therefore lowers the stable opportunity share for $SP$ workers. Note that $\gamma_E$ changes only the composition of accessible job offers; it does not alter the separation probabilities $\delta_S$ and $\delta_F$ associated with the realized stability types. The estimated values of the two family-structure wedges, $\alpha_U$ and $\gamma_E$, and their quantitative implications for the composition of accessible offers are presented in Section 5.2.

Regarding the relative separation risk of fragile and stable matches, I use the calibrated two-type structure in Baydur and Mukoyama (2020) as the external benchmark. Although their empirical hazard analysis uses the NLSY79, the stability type-specific separation probabilities in their quantitative model are calibration objects rather than estimates of latent stability types from the NLSY79. Their average monthly separation probabilities for fragile and stable types are set to 0.021 and 0.011, respectively, with the aggregate EU flow disciplined by aggregate U.S. worker-flow evidence. I therefore do not interpret these numbers as cohort-specific estimates.

Because the our model operates at a weekly frequency, I convert the two monthly probabilities to weekly equivalents by preserving monthly survival:

$$1 - \delta_s^M = (1 - \delta_s^W)^{52/12}, \qquad s \in \{F, S\}.$$

The resulting weekly-equivalent relative separation-risk benchmark is

$$r_\delta = \frac{1 - (1 - 0.021)^{12/52}}{1 - (1 - 0.011)^{12/52}} = 1.91654.$$

Only this relative separation risk spread is taken externally. The weekly stable-type separation probability $\delta_S$ is estimated jointly with the remaining structural parameters from the NLSY97, and the fragile-type probability is implied by $\delta_F = r_\delta \delta_S$.

I set the annual discount factor 0.96 and convert it to the weekly frequency. The benchmark unemployment payoff is set to $b = 1$ on the normalized payoff scale; Section **??** varies the $SP$ unemployment payoff directly.

Table 8 summarizes the quantities held fixed outside the joint minimum-distance estimation. The relative wage nodes and the dispersion of nonwage match value are disciplined by the NLSY97, whereas the remaining quantities are external inputs, benchmark assumptions, normalizations, or quantities implied by them. In particular, the unemployment resource level is normalized to $b = 1$, and the empirical relative wage nodes are rescaled into the employment-resource nodes used in the log utility specification.

### 5.2 Estimation and Targeted Fit

The ten free parameters are jointly estimated by minimum distance:

$$\theta = (\lambda_U, \lambda_E, \eta_{U,L}, \eta_{U,H}, \eta_{E,L}, \eta_{E,H}, \rho_E, \delta_S, \alpha_U, \gamma_E)'.$$

Let $\widehat{m}_j$ denote the empirical target $j$, $m_j(\theta)$ its model counterpart, and $s_j$ the empirical scale used to standardized the discrepancy. The minimum-distance objective is

$$\widehat{\theta} = \arg\min_{\theta \in \Theta} Q(\theta),$$

where

$$Q(\theta) = \sum_{j=1}^{J} d_j \left[ \frac{m_j(\theta) - \widehat{m}_j}{s_j} \right]^2,$$

where $d_j$ denotes the block weight.

The eleven core targets are chosen to discipline distinct parts of the model: the weekly UE and direct EE probabilities pooled across family-structure groups; the $TP$ weekly EU probability; the low and high quality shares among $TP$ jobs reached through direct job-to-job transitions; the

Table 8: Calibrated Inputs, Benchmark Assumptions, and Normalizations

| Quantity | Value | Source or quantitative role |
|---|---|---|
| ***Panel A. Data-based inputs held fixed outside the estimator*** | | |
| $\sigma_x$ | 0.99374 | NLSY97 directional wage-quality changes among direct EE moves |
| $(w_L, w_M, w_H)$ | (0.61157, 1, 2.12938) | NLSY97 residualized starting-wage distribution; middle node normalized to 1 |
| ***Panel B. Benchmark and external inputs*** | | |
| $\omega = b/w_M$ | 0.425 | U.S. Department of Labor unemployment-insurance data; benchmark ratio of unemployment resources to middle wage |
| $r_\delta = \delta_F/\delta_S$ | 1.91654 | Baydur and Mukoyama (2020) benchmark, converted to weekly frequency |
| $\nu_0$ | 0.50 | Equal stable and fragile baseline offer shares |
| $\beta$ | $0.96^{1/52} = 0.999215$ | Weekly equivalent of an annual discount factor of 0.96 |
| ***Panel C. Normalizations and implied resource scale*** | | |
| $b$ | 1 | Unemployment-resource normalization |
| $\eta_{U,M}, \eta_{E,M}$ | 0 | Reference category for wage-quality log weights |
| $\mathbb{E}[x]$ | 0 | Location normalization for nonwage match value |
| $(\widetilde{w}_L, \widetilde{w}_M, \widetilde{w}_H)$ | (1.43900, 2.35294, 5.01031) | Employment-resource nodes implied by $\widetilde{w}_q = w_q/\omega$ |

*Notes:* Data-based inputs in Panel A are constructed from the NLSY97 but are held fixed outside the joint minimum-distance estimation. The benchmark utility specification is $u(c) = \log c$, so the normalization $b = 1$ implies $u(b) = 0$; the resource level itself is 1, not zero. The employment-resource nodes are the quantities entering $u(\widetilde{w}_q) + x$.

quality-standardized $SP/TP$ EU hazard ratio; the low and high quality shares among $SP$ and $TP$ jobs entered from unemployment; and the same-quality share of direct job-to-job transitions. Here, pooled means that $TP$ and $SP$ workers are combined for the corresponding transition target. The model counterpart averages the group-specific transition probabilities using the empirical shares of $TP$ and $SP$ workers in the relevant unemployment or employment risk set.

Table 9 reports the ten parameters jointly estimated by minimum distance. The first eight parameters govern common offer arrival, wage-quality, persistence, and separation margins. The final two, $\alpha_U$ and $\gamma_E$, are the family-specific offer-access wedges. Although all parameters are estimated jointly, the baseline parameter is anchored first from $TP$ workers, and then the family-specific wedge parameters govern the deviation from the baseline distribution. To be more specific, $\eta_{o,L}, \eta_{o,H}$ are primarily disciplined by the level of accepted wage-quality among $TP$ workers, whereas $\alpha_U$ is disciplined by the additional $SP$–$TP$ difference in accepted wage quality among jobs entered from unemployment. Similarly, for match stability, the baseline stable-offer share is assumed to be $\nu_0 = 0$ for $TP$ workers as a benchmark distribution, while $\gamma_E$ is disciplined by

Table 9: Estimated Parameters and Their Empirical Discipline

| Parameter | Estimate | Interpretation and empirical discipline |
|---|---|---|
| $\lambda_U$ | 2.9639% | Common weekly job-offer arrival probability while unemployed; disciplined by the pooled UE transition probability through endogenous offer acceptance |
| $\lambda_E$ | 5.3114% | Common weekly job-offer arrival probability while employed; disciplined by the pooled direct EE transition probability through current-match survival and endogenous offer acceptance |
| $\eta_{U,L}$ | 0.386306 | U-origin log weight on low-quality offers relative to the middle-quality node; disciplined primarily by the low-quality share among $TP$ jobs reached from unemployment |
| $\eta_{U,H}$ | $-0.325769$ | U-origin log weight on high-quality offers relative to the middle-quality node; disciplined primarily by the high-quality share among $TP$ jobs from unemployment |
| $\eta_{E,L}$ | 1.153864 | E-origin log weight on low-quality offers relative to the middle-quality node; disciplined primarily by the low-quality share among $TP$ jobs reached through direct job-to-job transitions |
| $\eta_{E,H}$ | $-0759290$ | E-origin log weight on high-quality offers relative to the middle-quality node; disciplined primarily by the low-quality share among $TP$ jobs reached through direct job-to-job transitions |
| $\rho_E$ | 0.526928 | Persistence of wage quality in offers received while employed; disciplined by the same-quality share of direct job-to-job transitions |
| $\delta_S$ | 0.1348% | Weekly separation probability of a stable match; disciplined by the $TP$ EU transition probability through the stationary composition of stable and fragile matches |
| $\alpha_U$ | $-0.068633$ | $SP$ shift in the wage-quality composition of offers received while unemployed; disciplined by $SP$–$TP$ differences in the wage-quality composition of jobs entered from unemployment |
| $\gamma_E$ | $-0.925624$ | $SP$ shift in the log odds of receiving a stable rather than fragile offer while employed; disciplined by the quality-standardized $SP/TP$ EU hazard ratio for jobs reached through direct job-to-job transitions |

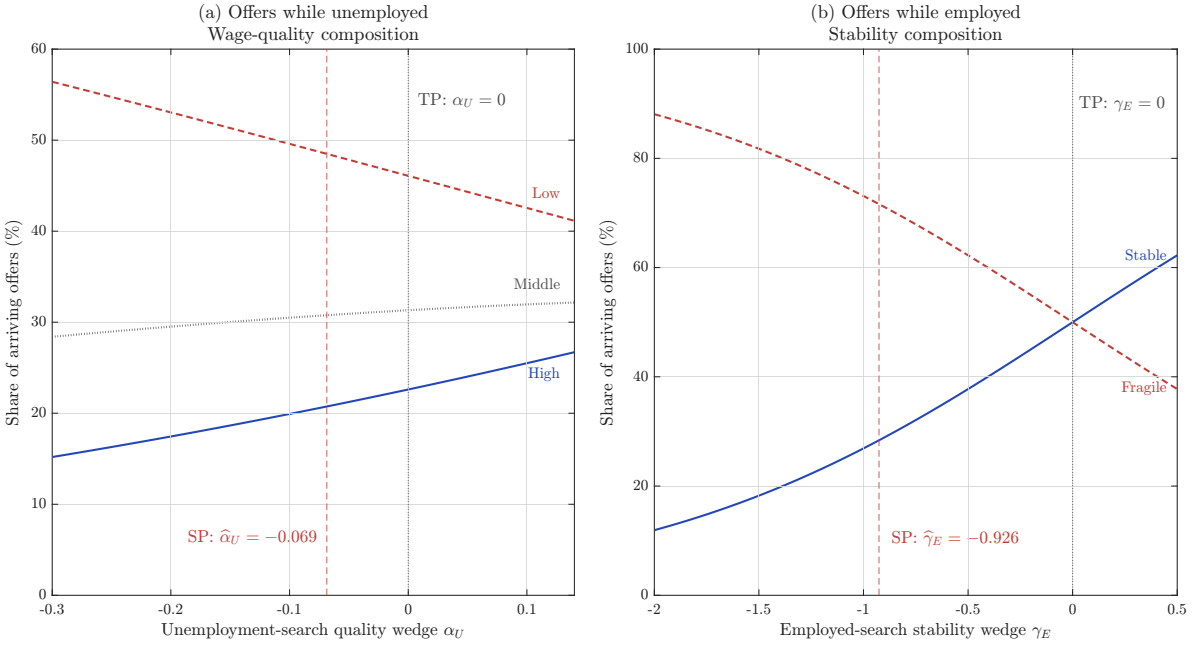
*Notes:* The parameters $\lambda_U$ and $\lambda_E$ are primitive job-offer arrival probabilities, not UE and EE transition probabilities. A UE transition also requires offer acceptance, while a direct EE transition requires current-match survival and offer acceptance. Similarly, $\delta_S$ is the separation probability conditional on a stable match rather than the aggregate $TP$ EU probability. The pooled UE and direct EE targets combine $TP$ and $SP$ workers using their empirical shares in the relevant unemployment and employment risk sets. All probabilities are weekly.

the remaining $SP/TP$ EU hazard ratio among E-origin jobs after standardizing their wage-quality composition.

Figure 2 and 3 translate these into estimates into their implications for the composition of accessible offers. The estimated family-specific margins are

$$\widehat{\alpha}_U = -0.069, \qquad \widehat{\gamma}_E = -0.926.$$

Figure 2: Family Structure and Accessible Job-Offer Distributions




*Notes:* Panel (a) shows how the U-origin quality wedge $\alpha_U$ shifts the conditional distribution of arriving offers across low-, middle-, and high-quality categories. The $TP$ benchmark is normalized to $\alpha_U = 0$, while the estimated $SP$ wedge is $\widehat{\alpha}_U = -0.069$. Panel (b) shows how the E-origin stability wedge $\gamma_E$ changes the conditional shares of stable and fragile offers. The $TP$ benchmark satisfies $\lambda_{E,S}^{TP} = \nu_0 = 0.5$, corresponding to $\gamma_E = 0$, while the estimated $SP$ wedge is $\widehat{\gamma}_E = -0.926$. The figure describes accessible-offer distributions conditional on an offer arriving, rather than accepted-job distributions.

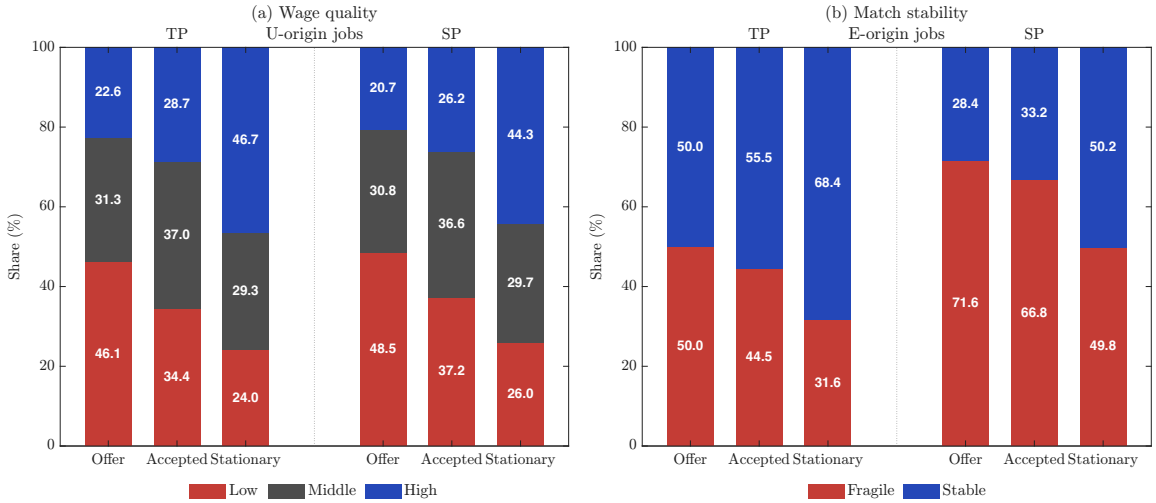
The first shifts $SP$ unemployment-search offers towards lower wage quality. Under the calibration $\nu_0 = 0.5$, the second implies

$$\widehat{\nu}^{SP} = \Pr(S \mid E, SP) = \Lambda(\widehat{\gamma}_E) \simeq 0.284,$$

compared with 0.5 for $TP$ employed search. This stable share is a latent model-implied object conditional on the benchmark stability structure, not a directly observed probability.

Table 10 compares the empirical targets with their model counterparts evaluated at $\widehat{\theta}$. The minimum-distance objective evaluated at the estimated parameter vector is $Q(\widehat{\theta}) = 0.222$. All eleven targeted model-data discrepancies are smaller than on empirical error scale in absolute value. The model closely reproduces pooled UE and EE transition probabilities, the $TP$ EU probability, the persistence of wage quality across direct EE moves, and the quantity-standardized E-origin family difference in employment exits. Note that the reported job-quality shares describe the composition of *realized* jobs rather than the composition of all offers received.

Figure 3: Composition of Job Offers, Acceptance, and Stationary Distributions by Family Structure




*Notes:* Panel (a) and (b) present the composition of offer distribution, job acceptance, and stationary distribution by family structure among jobs reached from unemployment and employment, respectively. Offer distributions are generated and implied by the model conditional on an offer arriving. The $TP$ distribution is evaluated at the normalization $\alpha_U = 0$ for the U-origin jobs and at $\gamma_E = 0$ for the E-origin jobs, while the corresponding $SP$ distribution is evaluated at $\widehat{\alpha}_U = -0.069$ and $\widehat{\gamma}_E = -0.926$. Accepted distributions additionally reflect endogenous workers' decisions. Stationary distributions are conditional on employment and incorporate repeated acceptance, separation, reemployment, and job-to-job mobility.

Table 10: Targeted Model Fit

| **Target moment** | **Data** | **Model** |
|---|---|---|
| Pooled weekly UE transition probability | 2.3421% | 2.3414% |
| Pooled weekly direct EE transition probability | 0.3677% | 0.3679% |
| $TP$ weekly EU transition probability | 0.1742% | 0.1738% |
| $TP$ job-to-job entries: low-quality share | 0.284110 | 0.284151 |
| $TP$ job-to-job entries: high-quality share | 0.401669 | 0.401302 |
| Job-to-job entries: quality-standardized $SP/TP$ EU hazard ratio | 1.162988 | 1.159077 |
| Accepted U-origin jobs, $SP$: low-quality share | 0.364541 | 0.372292 |
| Accepted U-origin jobs, $TP$: low-quality share | 0.348643 | 0.343638 |
| Accepted U-origin jobs, $SP$: high-quality share | 0.255748 | 0.261956 |
| Accepted U-origin jobs, $TP$: high-quality share | 0.290495 | 0.286616 |
| Direct job-to-job transitions: same-quality share | 0.532139 | 0.531295 |

*Notes:* The pooled UE and direct EE moments combine $TP$ and $SP$ workers using their empirical shares in the relevant risk set. The quality-standardized family difference is displayed as a hazard ratio for consistency with the empirical tables, but its estimation target is computed on the log-hazard-ratio scale. Transition probabilities are weekly, and probabilities and shares are reported as fractions. The minimum-distance criterion at the reported estimate is $Q(\widehat{\theta}) = 0.222$; it is a measure of model fit rather than a formal overidentification statistic.

### 5.3 Untargeted Moments Validation

I next examine whether the estimated model reproduces aggregate labor market outcomes beyond those directly fitted by the minimum-distance criterion. Panel A of Table 11 compares unemployment rates. The model generates stationary unemployment rates of 6.92 percent for $TP$ workers and 7.72 percent for $SP$ workers, implying $SP - TP$ gap of 0.8 percentage points. The corresponding empirical benchmark unemployment rates are 6.31 and 6.91 percent, with a gap of 0.61 percentage point. Thus, the model reproduces the direction and broad magnitude of the unemployment difference, although it overstates the gap by approximately 0.19 percentage point.

Panel B provides a further validation using weekly EU transitions. The $TP$ weekly EU probability is included among the estimation targets and therefore serve as the baseline anchor. The aggregate $SP$ probability and the aggregate $SP/TP$ hazard ratio, however, are not targeted; the family-specific EU target is instead the job-quality-standardized $SP/TP$ hazard ratio among jobs reached through direct job-to-job transitions. The model generates a weekly EU probability of 0.1963 percent for $SP$ workers, essentially identical to the empirical value of 0.1963 percent in the data. Applying the complementary log-log transformation used in the empirical analysis, the model probabilities imply an aggregate $SP/TP$ hazard ratio of 1.129, compared with 1.105 in Table 4. Thus, the model implies that $SP$ workers face a 12.9 percent higher aggregate EU hazard, compared with 10.5 percent in the data. Although the model somewhat overstates the relative difference, it reproduces its direction and approximate magnitude. This comparison provides an untargeted check that the estimated offer-access wedges and the resulting stationary match composition translate into the observed difference in aggregate employment instability.

## 6 Quantitative Implications of the Model

### 6.1 Mechanism Decomposition

The model generates stationary unemployment rates of 6.92 percent for $TP$ workers and 7.72 percent for $SP$ workers, a gap of 0.8 percentage points. The corresponding weekly EU probabilities are 0.1738% and 0.1963%.

I next conduct a mechanism decomposition to separate the roles of the two family-specific offer-access margins. For each outcome, I recompute the stationary equilibrium after setting $\alpha_U = 0$, setting $\gamma_E = 0$, and setting both wedges to zero. Because the model is nonlinear, the contribution assigned to one wedge can depend on whether the other wedge has already been removed. I therefore average each wedge's marginal contribution over the two possible equalization orders, following Shapley decomposition (Shapley et al., 1953; Shorrocks et al., 2013). Appendix C provides the formal construction. These decomposition exercises diagnose the sources of model-generated gaps rather than the effects of feasible policies: setting a wedge to zero mechanically removes the primitive family difference represented by that wedge.

Table 11: Validation of Aggregate Labor-Market Outcomes

| Outcome | Data benchmark | Model |
|---|---|---|
| ***Panel A. Unemployment rate*** | | |
| $TP$ workers | 6.31% | 6.92% |
| $SP$ workers | 6.91% | 7.72% |
| $SP - TP$ gap | 0.61 pp | 0.80 pp |
| ***Panel B. Weekly employment-to-unemployment transitions*** | | |
| $TP$ workers | 0.1742% | 0.1738% |
| $SP$ workers | 0.1963% | 0.1963% |
| $SP/TP$ EU hazard ratio | 1.105 | 1.129 |

*Notes:* $TP$ and $SP$ denote workers from two-parent and single-parent family households, respectively. In Panel A, the data benchmarks are standardized predicted unemployment probabilities from the fully controlled empirical specification, averaged over a common covariate distribution. The model column reports the unemployment mass in each group's stationary distribution. Stationary unemployment is not targeted in estimation. In Panel B, probability levels are expressed in percent per week. The empirical hazard ratio is the exponentiated $SP$ coefficient from the fully controlled complementary log-log specification reported in Table 4. The model hazard ratio is computed from the aggregate weekly transition probabilities as $[-\log(1 - h_{EU}^{SP})]/[-\log(1 - h_{EU}^{TP})]$. Because the empirical hazard ratio is a regression-based conditional object, it need not equal the ratio of the displayed standardized probability levels. The $TP$ weekly EU probability is an estimation target; the aggregate $SP$ probability and aggregate $SP/TP$ hazard ratio are untargeted validation outcomes. Differences may not equal differences between displayed rounded values.

Panel A of Table 12 shows that the employed-search stability wedge accounts for nearly all of the model-generated differences in aggregate employment instability. The benchmark model generates an aggregate weekly $SP$–$TP$ EU probability gap of 0.0225 percentage point, corresponding to an $SP/TP$ hazard ratio of 1.129. The employed-search stability wedge accounts for 99.38 percent of this gap. The unemployment-search quality wedge contributes 0.62 percent.

The stability margin also accounts for most of the stationary unemployment difference. The benchmark model produces an $SP$–$TP$ stationary unemployment gap of 0.798 percentage points. The stability wedge contributes 119.61 percent of this gap, whereas the quality wedge contributes $-19.61$ percent. The contribution above 100 percent means that the stability wedge by itself would generate a slightly larger unemployment gap and that the quality wedge modestly offsets it. Thus, the model attributes both higher aggregate employment-exit risk and higher stationary unemployment primarily to reduced access to stable opportunities while employed.

Panel B presents the long-run propagation result by decomposing the stationary job-quality composition of employment. In the stationary employed population, the model generates an $SP$–$TP$ gap of 1.96 percentage points in low-quality wage share and the corresponding gap of $-2.38$ percentage points in high-quality wage share. The wage quality wedge from U-origin jobs accounts for 54.67 percent of the low-quality gap and 44.87 percent of the high-quality gap, while the E-origin stability wedge accounts for the remaining 45.33 and 55.13 percent, respectively.

Table 12: Mechanism Decomposition of Family-Structure Gaps

| Outcome | *SP*–*TP* gap | Contribution (% of gap) | |
|---|---|---|---|
| | | Quality wedge ($\alpha_U$) | Stability wedge ($\gamma_E$) |
| ***Panel A. Aggregate labor-market outcomes*** | | | |
| Weekly EU transition probability | 0.0225 pp | 0.62 | 99.38 |
| Stationary unemployment rate | 0.7983 pp | −19.61 | 119.61 |
| ***Panel B. Stationary employment composition*** | | | |
| Low-quality employment share | 1.96217 pp | 54.67 | 45.33 |
| High-quality employment share | −2.3774 pp | 44.87 | 55.13 |

*Notes:* Gaps are *SP* minus *TP* and are measured in percentage points. The last two columns report order-averaged Shapley contributions obtained by equalizing $\alpha_U$ and $\gamma_E$ separately and jointly. The weekly EU probability gap corresponds to an *SP*/*TP* hazard ratio of 1.129. None of the reported gaps is directly targeted. Negative contributions indicate offsets; contributions above 100 percent arise when the other wedge moves the outcome in the opposite direction. These mechanism decompositions provide model accounting rather than policy effects.

## 6.2 Welfare Analysis

We have seen that differences in job-offer distributions over wage quality and match stability can generate differences in EU transition rates and unemployment rates across family structures. In this section, I measure the welfare implications of these differences in consumption-equivalent terms. In other words, I ask how much of welfare loss of *SP* workers would be when employment instability caused by different distributions of jobs available to them is converted into pecuniary sense.

In order to measure the welfare loss, I consider a counterfactual economy in which *an SP* worker faces the same job-offer distributions as *TP* worker in both unemployed and employed search, by setting $\alpha_U = 0$ and $\gamma_E = 0$ for the *SP* worker. Recall that $V_o^{SP}$ denote the *SP* type worker's expected discounted lifetime utility under the baseline economy, conditional on initial labor market state $o \in \{U, E\}$. Let $W_o^{SP}$ denote the corresponding value under a counterfactual economy. To evaluate the two economies from a same starting point, I use the baseline *SP* stationary distribution, $\mu^{SP}(z)$, $z \in \{U, m\}$, as the distribution of initial state.

Assume that consumption in a given state is $c$ and it is compensated to be $\kappa_o$ times in each period, depending on search state $o$. Then, the increase in utility for that state is

$$\log[(1+\kappa_o)c] - \log c = \log(1+\kappa_o).$$

If an amount of utility equal to $\log(1+\kappa)$ is added each period permanently, its present value is

$$\sum_{t=0}^{\infty} \beta^t \log(1+\kappa_o) = \frac{\log(1+\kappa_o)}{1-\beta}.$$

Therefore, the compensated $SP$ worker's lifetime utility is

$$W_o^{SP} = V_o^{SP} + \frac{\log(1+\kappa_o)}{1-\beta},$$

which implies

$$\kappa_o = \exp\left[(1-\beta)W_o^{SP} - V_o^{SP}\right] - 1.$$

For each search state $o$ and match characteristics $m$, We can calculate a consumption-equivalent compensation multiplier, and by taking a weighted average of these values according to the stationary distribution, we can determine a common compensation multiplier $\kappa$ for $SP$ type workers, regardless of search state.

Given that the model is true with estimated family-structure wedge parameter, this welfare analysis gives $\kappa = 5.83$, which implies that a uniform permanent increase of 5.83% in consumption to $SP$ workers is needed to equalize to $TP$ workers' welfare. Note that the time unit is weekly in the model, so the model-implied welfare loss of $SP$ workers is considerable.

## 6.3 Policy Implications

The preceding subsection documents that consumption-equivalent welfare loss captured by employment instability of $SP$ type workers are not negligible. Family structure during adolescence cannot be changed retrospectively and it is exogenous variable outside of model. However, the labor market opportunities associated with family structure may be amenable by policy intervention with proper policy tools. In this section, we examine actionable policy to reduce employment instability by family structure and evaluate effectiveness of policy. We investigate existing labor policies for $SP$ workers and discuss how they can improve employment stability across family structures.

### 6.3.1 ASEP Pulic-Sector Recruitment Policy in Greece

A labor policy explicitly targeting $SP$ workers can be found in Greece's public-sector recruitment system. ASEP (Supreme Council for Civil Personnel Selection) is an independent recruiting authority in Greece, established in 1994 for transparency, publicity, objectivity, and meritocracy in the civil personnel hiring process. ASEP guarantees that applicants qualifying as children of single-parent households receive 100 additional ranking points for positions in the compulsory education qualification category (YE). The additional 100 points can provide a meaningful advantage, particularly for applicants with limited work experience. Under the standard scoring rules shown in the February 2026 Patmos municipality recruitment notice, the 100-point child-of-single-parent-household eligibility is equivalent to 14.3 months of work experience, which is more than a master's degree attainment; holding a master's degree is evaluated as 70 additional points.

The benefit is not necessarily limited to applicants currently living with a single parent. The

eligibility condition relevant to family circumstances refers to parental responsibility exercised for a period *before* the applicant reached adulthood. For the child-of-single-parent category, the January 2026 documentation describes conditions that occurred before adulthood. The provision of advantage points does not necessarily guarantee employment, but the policy is one of examples that recognizes single-parent household as a basis for improving access to employment.

This example can be interpreted as opportunity expansion counterfactuals in the model context. The provision of extra points to $SP$ workers may affect both the frequency and the composition of job offers available to $SP$ workers. In the model, higher access to higher-wage jobs during unemployed search would narrow the wage-quality wedge $\alpha_U$, while higher access to stable offers during employed search would narrow the stability wedge $\gamma_E$. The provision of extra points to $SP$ workers may increase the frequency of job arrival for $SP$ workers $\lambda_U^{SP}, \lambda_E^{SP}$. Richer data about policy results will enable us to identify which parameter this policy can affect.

#### 6.3.2 WIOA Youth Program in Prince George's County, Maryland

We can also find a similar labor policy targeting $SP$ workers in the United States. WIOA (Workforce Innovation and Opportunity Act) Youth program in Prince George's County in Maryland state explicitly recognizes *currently* living in a single-parent household as an employment barrier. Applicants with eligibility can be provided assistance to complete education or secure, or retain employment. WIOA Youth program provides occupational skills training, mentoring, counseling, and continuous follow-up support even after employment. The eligibility condition depends only on *current* family structure, not on the adolescent period, but it illustrates how family circumstances can inform the provision of education and employment assistance during the transition to adulthood and the transition from school to work. Evidence from broader youth employment programs suggest that WIOA Youth Program can improve short-run labor market outcomes, but no significant effect for longer-run period. Whether such policies reduce employment stability gap associated with family structure remains an open empirical question.

## 7 Conclusion

This paper studies how family structure during adolescence is associated with employment instability in young adulthood. Using weekly labor market histories from the NLSY97, I show that the higher post-school unemployment experienced by workers from single-parent households is driven primarily by more frequent employment-to-unemployment transition rather than by slower unemployment-to-employment transition rates. The evidence points to two distinct margins along the job ladder: $SP$ workers enter a less favorable distribution of jobs following unemployment, and they experience lower subsequent match durability among jobs reached through direct job-to-job transitions, even after accounting for observed job quality. A quantitative job-ladder model represents these patterns as differences in the quality composition of opportunities

accessible from unemployment and the stability composition of opportunities accessible while employed. The model attributes most of the aggregate EU transition hazard ratio and stationary unemployment gaps to the stability margin. The welfare loss due to employment instability caused by different job-offer distributions is equivalent to persistent 5.8% less consumption for *SP* workers. This welfare loss calls for relevant policy intervention.

The paper makes three main contributions. First, it brings a flow-based perspective to the relationship between family structure and young-adult labor market outcomes. Rather than treating unemployment as a static outcome, the analysis identifies the transitions through which family-structure differences accumulate. Second, it shows that family-structure differences extend *beyond* educational attainment and the probability of finding a job. The composition and subsequent durability of accepted matches provide additional dimensions of early-career inequality. In particular, distinguishing jobs entered from unemployment from those reached through direct job-to-job transitions reveals different family-structure patterns that would be obscured in a pooled sample of accepted jobs. Third, the quantitative framework separates two proximate margins of opportunity access and traces how they interact through acceptance, separation, reemployment, and job-to-job mobility. This provides an accounting framework for connecting entry conditions to longer-run differences in employment continuity.

There are several limitations of the analysis. The NLSY97 observes accepted jobs rather than rejected offers. The estimated wedges therefore combine potentially distinct sources of limited access and should not be interpreted as literal firm-side offer distributions. Finally, the model implications hold the estimated environment fixed and do not incorporate firm responses, general-equilibrium effects, or parameter uncertainty. They should be read as model-generated benchmarks for comparing margins rather than as estimates of the causal effects of particular programs. These limitations point to a natural direction for future research: identifying the underlying factors that generate unequal access to high-quality and durable matches. Suggestive evidence provided in Appendix B hints at time-transfer channel of family structure and richer data and further research is required for the rigorous identification.

In conclusion, in spite of several limitations, the research results require us to look beyond whether and how quickly young workers find employment in order to understand early-career inequality in depth. The types of jobs they can effectively access, and whether those jobs provide durable employment, may be equally important in terms of employment stability and close labor market inequality across family structures.

# Appendix A Observed Noncognitive Characteristics

The main results from Table 3 leaves a remaining association between family structure and unemployment after conditioning on observed education and family financial resources. One possible concern is that this association may partly reflect persistent worker characteristics that are not included in the main specification. I examine this possibility using the noncognitive measures available in the NLSY97.

**Measure construction** I use two sets of noncognitive measures from the NLSY97. The first is based on eight Goldberg personality items collected in Round 6. These items describe respondents along dimensions including organization, conscientiousness, dependability, thoroughness, agreeableness, cooperativeness, flexibility, and trustfulness. I recode these items so that higher values consistently indicate more of the relevant trait and then I combine them into measures of conscientiousness and agreeableness. Each measure is converted into a $z$-score by subtracting its sample mean and dividing by its sample standard deviation.

The second measure is an earlier respondent-reported behavioral and emotional problems scale from Round 1. Higher values of this measure indicate more behavioral or emotional problems. This measure is distinct from the Goldberg personality items and was collected substantially earlier in the respondents' lives.

The measures used here is different from those in Cunha et al. (2006), whose NLSY79 analysis identifies a noncognitive factor using the Rosenberg Self-Esteem and Rotter Locus-of-Control scales. My objective is not to replicate their latent-factor measure, but to use the earlier noncognitive characteristics available in the NLSY97 as proxies for dimensions of persistent worker heterogeneity that may be relevant for labor market outcomes.

The Goldberg items were collected in 2002–2003 and are available only for the younger among NLSY97 birth cohorts. To ensure that the personality measures *precede* the labor market outcomes used in this exercise, I restrict the outcome window to 2006 and later. This restriction, together with the limited cohort coverage of the Goldberg module, substantially reduces the number of respondents from 7347 in the main analysis sample to 4029 in the Goldberg sample.

**Empirical design** I re-estimate the post-school labor force unemployment specification used in the main analysis. The dependent variable equals 1 in an unemployed week and 0 in an employed week. All specifications include the full set of controls used in the most controlled unemployment specification in the main text: demographic, own education, parental education, calendar quarter fixed effects, and family financial resources.

Because the Goldberg measures are observed only for a restricted subset of respondents, I separate changes induced by sample restriction from changes induced by adding the noncognitive controls. I first compare the average marginal effect (AME) of single-parent household in the main analysis sample with the corresponding estimate in the Goldberg sample. I directly test whether

the estimated family-structure association differs from the main and Goldberg sample. Because the Goldberg sample is a subset of the main sample, the comparison accounts for the fact that many of the same respondents contribute to both estimates. I implement this comparison by estimating the two specifications jointly and clustering inference at the respondent level.

I then hold the estimation sample fixed and compare the family-structure estimate before and after adding the noncognitive measures. The first comparison adds conscientiousness and agreeableness within the Goldberg sample. The second additionally includes behavioral and emotional problems and restricts both the baseline and augmented specifications to the 4016 respondents for whom all three measures are observed. These same-sample comparison separate the change induced by adding the controls from the change induced by restricting the sample.

**Results and interpretation** Panel A of Table B.1 reports sample-sensitivity comparison. In the main sample, the estimated average marginal effect of single-parent household is 0.61 percentage points and is statistically significant at the 5 percent level. In the Goldberg sample, the estimate declines to 0.41 percentage points and is less precisely estimated. The difference between the two estimates is 0.20 percentage points, but the equality test does not reject the null that the two AMEs are the same ($p = 0.470$). Thus, although the estimate becomes smaller in the Goldberg sample, the results do not establish that the family-structure association becomes weaker among respondents in the Goldberg subsample.

Panel B reports the same-sample comparisons. The noncognitive measures themselves contain meaningful information about unemployment risk. For example, a one-standard-deviation increase in conscientiousness is associated with an approximately 0.48 percentage-point lower probability of unemployment and is statistically significant at the 5 percent level. The estimate for agreeableness is close to zero and statistically insignificant. A one-standard-deviation increase in behavioral and emotional problems is associated with an approximately 0.30 percentage-point higher unemployment probability and is statistically significant at the 10 percent level. The controls are jointly significant: the joint-test $p$-value is 0.028 for conscientiousness and agreeableness and 0.010 when behavioral and emotional problems are included.

Adding these controls reduce the estimated family-structure association. Within the Goldberg sample, adding conscientiousness and agreeableness reduces the single-parent AME from 0.41 to 0.37 percentage points, an attenuation of approximately 9 percent. In the common sample with all three measures, the estimate declines from 0.42 to 0.35 percentage points. This decline of 0.07 percentage points corresponds to an attenuation of approximately 16 percent. The attenuation is statistically significant at the 10 percent level ($p = 0.073$). This provides suggestive evidence that observed noncognitive characteristics account for part of the family-structure difference in unemployment.

These results are informative but should be interpreted with caution. The family-structure estimate is already imprecisely estimated in the restricted-sample baseline, *before* the noncognitive

Table A.1: Observed Noncognitive Characteristics and Post-School Unemployment

| | Sample comparison | | Same-sample comparison | |
|---|---|---|---|---|
| | Main | Goldberg | Baseline | + C/A/ Behavior |
| ***Panel A. Family-structure estimate*** | | | | |
| *SP–TP* difference in unemployment probability (percentage points) | 0.61** | 0.41 | 0.42 | 0.35 |
| | (0.29) | (0.37) | (0.37) | (0.36) |
| Difference in AMEs (percentage points) | 0.20 | | 0.07* | |
| Difference test, $p$-value | 0.476 | | 0.073 | |
| Attenuation | – | | 16.2% | |
| ***Panel B. Noncognitive characteristics*** | | | | |
| Conscientiousness | – | – | – | -0.48** |
| | | | | (0.20) |
| Agreeableness | – | – | – | -0.00 |
| | | | | (0.19) |
| Behavioral/emotional problems | – | – | – | 0.30* |
| | | | | (0.17) |
| Joint test of noncognitive controls, $p$-value | – | – | – | 0.010 |
| Respondents | 7,347 | 4,029 | 4,016 | 4,016 |
| Respondent-weeks | 4,821,227 | 2,330,505 | 2,323,628 | 2,323,628 |

*Notes:* The dependent variable equals 1 if the respondent is unemployed and 0 if employed in a given post-school labor-force week. Entries for the single-parent indicator and noncognitive characteristics report average discrete changes in predicted unemployment probabilities from binary logit models and are expressed in percentage points. The main and Goldberg estimates in the first two columns are evaluated on their respective estimation samples. The final two columns use the identical sample of respondents with complete information on conscientiousness, agreeableness, and behavioral or emotional problems. The reported differences equal the baseline AME minus the corresponding comparison AME. The first difference compares the main and Goldberg samples. The second compares the baseline and noncognitive-control specifications estimated on the same sample. Because the same respondents contribute to both estimates in each comparison, the difference tests account for their common information by estimating the relevant specifications jointly and clustering inference at the respondent level. The noncognitive measures are converted into z-scores with a mean of 0 and a standard deviation of 1. Their estimated effects therefore correspond to a one-standard-deviation increase in the relevant characteristic. All specifications include the controls from the fully controlled unemployment specification in the main text. Standard errors are clustered at the respondent level and reported in parentheses. The joint-test row reports the $p$-value for the null that the included noncognitive controls are jointly zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

controls are added. The comparison therefore does not show that adding these characteristics causes an otherwise precisely estimated association to disappear. Instead, it shows that conditioning on the observed noncognitive measures reduces the point estimate, with statistical evidence that is suggestive rather than conclusive.

The exercise has two additional limitations. First, the available NLSY97 measures capture only selected dimensions of noncognitive characteristics and cannot rule out other forms of persistent worker heterogeneity. Second, these characteristics may themselves partly reflect earlier family environments. Conditioning on them therefore does not identify a causal mediation effect

of noncognitive characteristics.

I interpret this exercise as suggestive evidence that noncognitive characteristics may account for a limited but non-negligible portion of the association between family structure and unemployment. The measures themselves predict unemployment, and their inclusion reduces the estimated family-structure difference by approximately 16 percent. However, the available data do not allow me to determine precisely how much of the gap is attributable to these characteristics or whether they represent an underlying mechanism. A more definitive analysis would require richer measure of noncognitive development observed at multiple stages of childhood and adolescence, together with information on family structures before these characteristics are formed. I therefore leave a more detailed investigation of this channel for future research when such data become available.

# Appendix B Grandparent Availability and Employment Instabiltiy Among Workers with Children

The main analysis shows that workers from single-parent households experience a higher EU hazard, even after conditioning on observed human capital, parental education, and family financial resources. The main text studies whether this instability is associated with the types of jobs workers enter and with subsequent match stability. This appendix considers a complementary worker-side explanation: employment continuity may also depend on workers' access to informal time and logistical support from family. To be more specific, I examine whether the availability of an additional grandparent is associated with employment stability among workers raising children. Grandparents may provide the time transfer and practical support through childcare, transportation, and assistance during unexpected employment disruptions. Living with two grandparents rather than one may therefore provide greater potential access to such resources.

**Empirical design** Because the empirical analysis in the main text focuses on family structure at age 16, I restrict the sample to workers whose adolescent family structure is consistent with their parental co-residence at the survey interview. Specifically, I consider employed respondents with at least one co-resident child younger than 18 and retain two groups: workers from single-parent households at 16 who currently live with one of their own parents ($SP$-concordant) and workers from two-parent households at age 16 who currently live with both parents ($TP$-concordant). This restriction aligns the contemporaneous parental co-residence with the family-structure distinction underlying the main analysis, allowing me to explore the potential role of an additional parent in providing support to workers with children.

I estimate a discrete-time complementary log-log model of EU transitions. The specification includes the same full set of controls used in Table 4 in the main text. The coefficient of interest compares the EU hazard of $SP$-concordant workers with that of $TP$-concordant workers.

Table B.1: Parental Co-residence and EU Hazards

| Specification | Workers | EU events | SP/TP HR | $p$-value |
|---|---|---|---|---|
| Full controls | 347 | 82 | 1.438<br>(0.336) | 0.120 |
| Full controls + youngest child age | 347 | 82 | 1.481<br>(0.356) | 0.102 |

*Notes:* The sample consists of employed respondents with at least one co-resident child younger than 18 whose parental configuration at the survey interview is concordant with their family structure at age 16. The SP-concordant group includes respondents from single-parent households at age 16 who currently live with one parent, whereas the TP-concordant group includes respondents from two-parent households at age 16 who currently live with both parents. The reported hazard ratio compares the EU hazard of SP-concordant workers with that of TP-concordant workers. All specifications include sex, race, birth-year and calendar-year fixed effects, own and parental education, family financial resources, school-exit resources, and log job tenure. Delta-method standard errors of the hazard ratios, clustered at the respondent level, are reported in parentheses.

**Results and interpretation** Table B.1 reports the relationship between parental co-residence and employment stability among workers with children whose adolescent and adult family households are concordant. In the fully controlled specification, the estimated EU hazard ratio is 1.438 ($p = 0.120$), indicating that $SP$-concordant workers experience an approximately 43.8 percent higher EU hazard than $TP$-concordant workers. Although the estimate is not statistically significant at conventional levels, its magnitude is *suggestive* of a potential role for an additional parent in helping employment stability. Adding the age of the youngest child yields a similar estimate (HR=1.481, $p$=0.102).

One possible interpretation is that an additional co-resident grandparent increases the household's capacity to provide time transfers, particularly childcare. When two grandparents are available to assist with childcare rather than one, workers may have greater flexibility in responding to unexpected childcare needs without interrupting employment. The estimated difference in EU hazards is consistent with this potential time-transfer channel, although this analysis does not directly observe the amount of childcare or other assistance provided by each grandparent. While this analysis does not identify as a causal effect of having two parents, it highlights a possible channel what family structure captures in terms of employment stability for workers after factoring out the effects of human capital and financial resources.

# Appendix C Construction of the Mechanism Decomposition

This appendix describes how I assign each model-generated family gap to the unemployment-search quality wedge, $\alpha_U$, and the employed-search stability wedge, $\gamma_E$. The procedure follows the Shalpey decomposition. It is useful because the model is nonlinear: the change produced by equalizing one wedge can depend on whether the other wedge has already been equalized.

For a given outcome, let $\Delta_{ab}$ denote the model-generated $SP$–$TP$ difference, where $a \in \{0, 1\}$

indicates whether the $\alpha_U$ is active and $b \in \{0,1\}$ indicates whether the $\gamma_E$ is active. The four model configurations are

$$\Delta_{11} = \Delta(\widehat{\alpha}_U, \widehat{\gamma}_E), \quad \Delta_{10} = \Delta(\widehat{\alpha}_U, 0),$$
$$\Delta_{01} = \Delta(0, \widehat{\gamma}_E), \quad \Delta_{00} = \Delta(0,0).$$

Thus, $\Delta_{11}$ is the difference generated by the estimated model, $\Delta_{10}$ retains only the quality wedge, $\Delta_{01}$ retains only the stability wedge, and $\Delta_{00}$ removes both wedges. Because the two family groups are otherwise identical in the model, $\Delta_{00} = 0$ for the outcomes considered here.

The contribution of the unemployment-search quality wedge is

$$\phi_\alpha = \frac{1}{2}\left[(\Delta_{11} - \Delta_{01}) + (\Delta_{10} - \Delta_{00})\right].$$

The first term measures the contribution of $\alpha_U$ when $\gamma_E$ remain active, while the second measures its contribution when $\gamma_E$ has already been removed.

Similarly, the contribution of the employed-search stability wedge is

$$\phi_\gamma = \frac{1}{2}\left[(\Delta_{11} - \Delta_{10}) + (\Delta_{01} - \Delta_{00})\right].$$

The two contributions satisfy the adding-up condition

$$\phi_\alpha + \phi_\gamma = \Delta_{11} - \Delta_{00}.$$

To make the role of the equalization order transparent, the two paths can also be written as

$$\Delta_{11} \xrightarrow{\alpha_U=0} \Delta_{01} \xrightarrow{\gamma_E=0} \Delta_{00}, \qquad \Delta_{11} \xrightarrow{\gamma_E=0} \Delta_{10} \xrightarrow{\alpha_U=0} \Delta_{00}.$$

For consistency with the Shapley formulas, an order-specific contribution is reported as the gap before equalization minus the gap after equalization. It is therefore the amount of gap attributed to the wedge at that step, not the after-minus-before change in the reported outcome.

Table C.1 reports both the gap remaining in each of the four model configurations and the marginal contribution obtained along each equalization path. The results illustrate why averaging over the two orders is useful. For example, equalizing $\alpha_U$ first raises the stationary unemployment gap from 0.798255 to 1.009222 percentage points, so its first-order contribution is $-0.210967$ point. If $\gamma_E$ has already been equalized, the corresponding contribution of $\alpha_U$ is $-0.102127$ point. Their average is $-0.156547$ point, or $-19.61$ percent of the full model gap, which is the Shapley contribution reported in the main text. Conversely. the stability wedge contributes 0.900382 point when it is removed first and 1.009222 points when it is removed second, for an order-averaged contribution of 0.954802 point, or 119.61 percent of the full gap.

Table C.1: Order-Specific Mechanism Decomposition

| | *SP–TP* gap under each model configuration | | | |
|---|---|---|---|---|
| Outcome | $\Delta_{11}$ Full model | $\Delta_{01}$ $\alpha_U = 0$ | $\Delta_{10}$ $\gamma_E = 0$ | $\Delta_{00}$ Both zero |
| Weekly EU probability (pp) | 0.022461 | 0.022135 | -0.000048 | 0 |
| Stationary unemployment rate (pp) | 0.798255 | 1.009222 | -0.102127 | 0 |
| Quality-standardized EU log hazard ratio, direct job-to-job entry | 0.147624 | 0.146782 | -0.000032 | 0 |
| Low-quality share, jobs entered from unemployment (pp) | 2.865416 | -1.4707654 | 3.610311 | 0 |
| High-quality share, jobs entered from unemployment (pp) | -2.465984 | 0.789322 | -2.776845 | 0 |
| Low-quality share, stationary employment (pp) | 1.962116 | 0.777676 | 0.961074 | 0 |
| High-quality share, stationary employment (pp) | -2.377430 | -1.130901 | -0.886810 | 0 |

| | Equalize $\alpha_U$ first | | Equalize $\gamma_E$ first | |
|---|---|---|---|---|
| Outcome | $\alpha_U$ first $\Delta_{11} - \Delta_{01}$ | $\gamma_E$ second $\Delta_{01} - \Delta_{00}$ | $\gamma_E$ first $\Delta_{11} - \Delta_{10}$ | $\alpha_U$ second $\Delta_{10} - \Delta_{00}$ |
| Weekly EU probability (pp) | 0.000326 | 0.022135 | 0.022509 | -0.000048 |
| Stationary unemployment rate (pp) | -0.210967 | 1.009222 | 0.900382 | -0.102127 |
| Quality-standardized EU log hazard ratio, direct job-to-job entry | 0.000842 | 0.146782 | 0.147656 | -0.000032 |
| Low-quality share, jobs entered from unemployment (pp) | 4.273070 | -1.407654 | -0.744895 | 3.610311 |
| High-quality share, jobs entered from unemployment (pp) | -3.255306 | 0.789322 | 0.310861 | -2.776845 |
| Low-quality share, stationary employment (pp) | 1.184440 | 0.777676 | 1.001042 | 0.961074 |
| High-quality share, stationary employment (pp) | -1.246529 | -1.130901 | -1.490620 | -0.886810 |

*Notes:* Each entry is an *SP–TP* difference. Probability and share differences are in percentage points; the quality-standardized EU hazard difference for jobs reached through direct job-to-job transitions is on the log-hazard-ratio scale. In the lower panel, a contribution is the gap before equalizing the indicated wedge minus the gap afterward. Thus, a negative entry means that the wedge offsets the full model-generated gap at that step. The Shapley contribution of each wedge is the average of its "first" and "second" entries. Small discrepancies can arise from rounding.

I report each contribution as a percentage of the full model-generated difference:

$$S_\alpha = 100 \times \frac{\phi_\alpha}{\Delta_{11} - \Delta_{00}}, \qquad S_\gamma = 100 \times \frac{\phi_\gamma}{\Delta_{11} - \Delta_{00}}.$$

A negative share indicates that a wedge offsets rather than widens the full difference. Consequently, when one wedge has a negative contribution, the contribution of the other wedge can exceed 100 percent while the two shares contribute to sum to 100 percent.

The scale of $\Delta_{ab}$ follows the outcome being decomposed. For stationary unemployment and

aggregate weekly EU transitions, it is the $SP$–$TP$ probability difference. For job-quality shares, it is the $SP$–$TP$ difference in percentage points. For the job-quality-standardized employment-exit hazard ratio among jobs reached through direct job-to-job transitions, the decomposition is performed on the log-hazard-ratio scale so that contributions are additive. All calculations are evaluated at the estimated parameter vector. They provide model accounting and do not incorporate parameter uncertainty or identify the causal effect of a particular policy intervention.

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